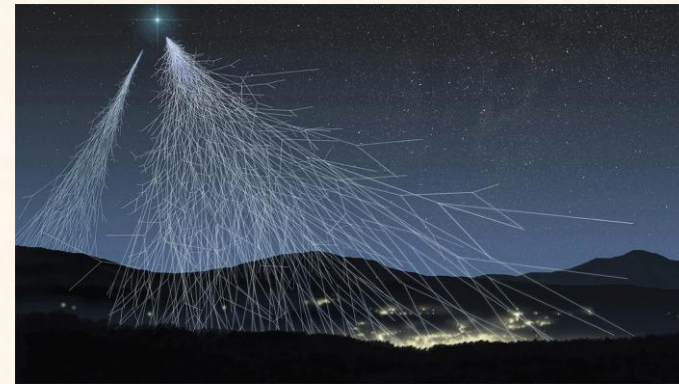
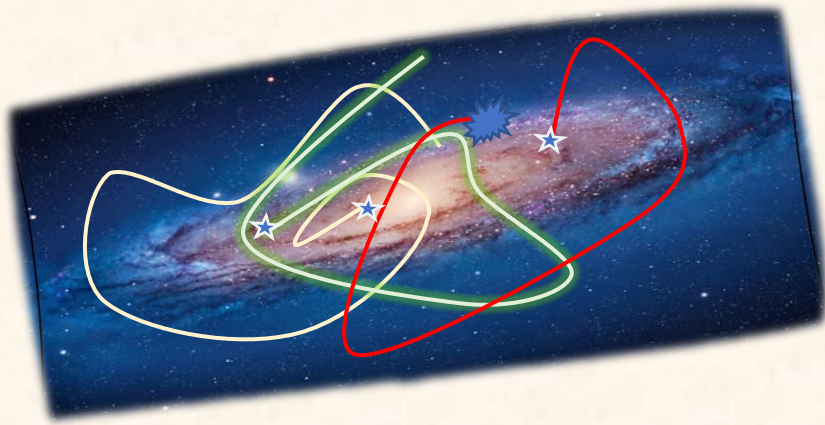


Introduction to Galactic Cosmic Rays



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Ramón y Cajal fellowship (UAM - Madrid)

Palestinian Advanced School - Cairo

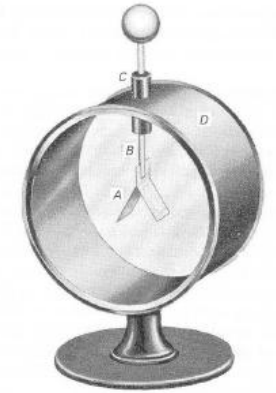
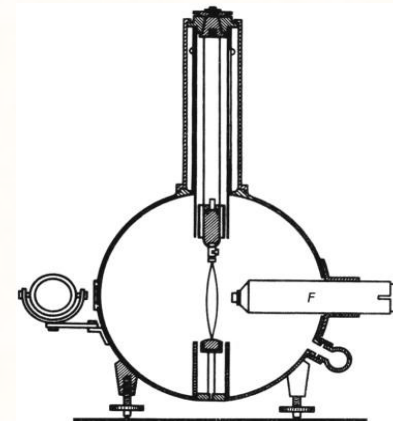
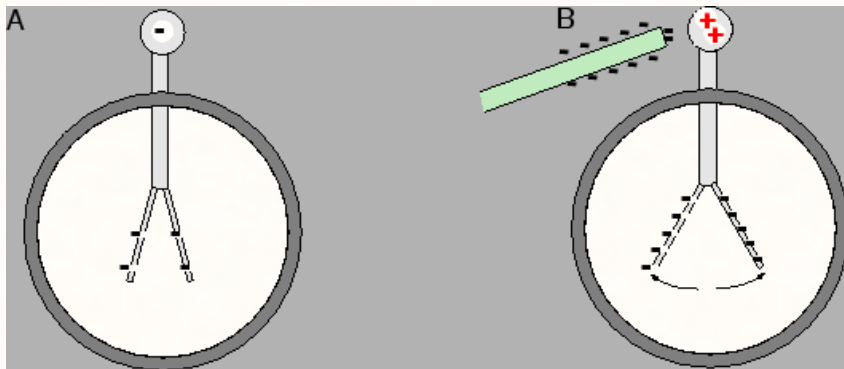
Outline

- ◆ What are Cosmic rays? What do we “observe”? and how?
- ◆ Where do they come from?
- ◆ Their movement in the Galaxy
- ◆ Interactions with the interstellar medium
- ◆ Production of secondary CRs

How we first “met” cosmic rays

The beginning

- Electroscopes showed a spontaneous discharge, even when they were insulated
 - Coulomb found that **electroscopes can spontaneously discharge** by the action of the air (1785)
 - Faraday confirms it, with better insulation technology (1835)
 - The speed of discharge decreases when air pressure is reduced (Crookes, 1879) → **Direct agent is the ionized air**
 - In the 1900s it was clear that in the **presence of a radioactive material** a charged electroscope quickly discharges

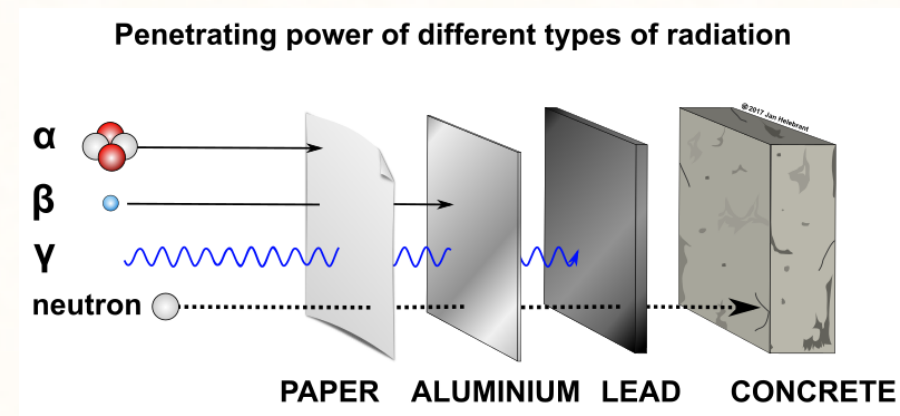
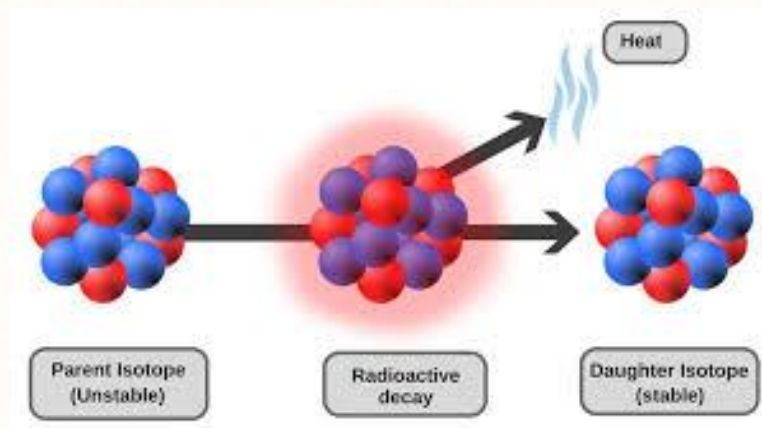
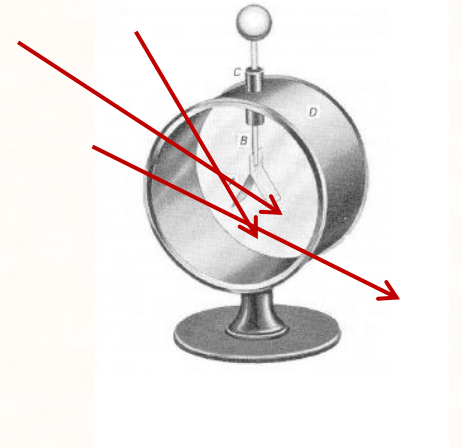


How we first “met” cosmic rays

The beginning

- Radiation charges the air, which redistribute the leaves' charges
Radioactive elements emit charged particles, that cause the discharge of the electroscopes. **The discharge rate of an electroscope was then used to measure the level of radioactivity**
- But where does this environmental radiation come from? People thought that charged particles came from the Earth's crust

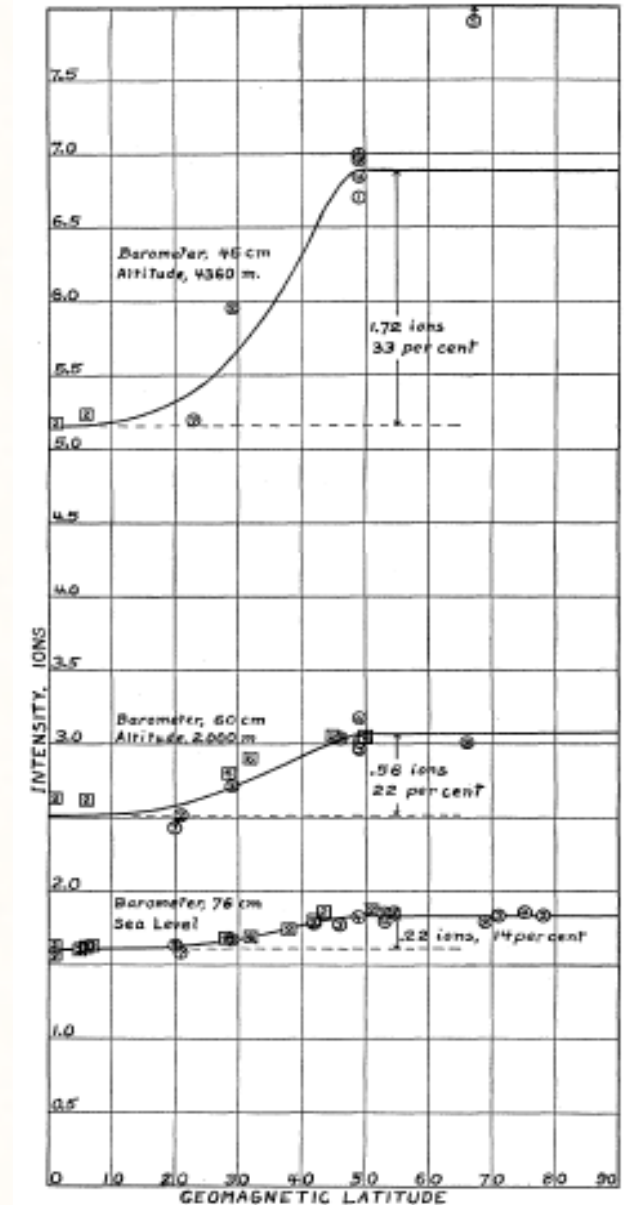
Observation: There is 28% decrease in discharge rate when the electroscope is taken from the surface down to the bottom of a salt mine (Elster & Geitel, 1908)



How we first “met” cosmic rays

The beginning

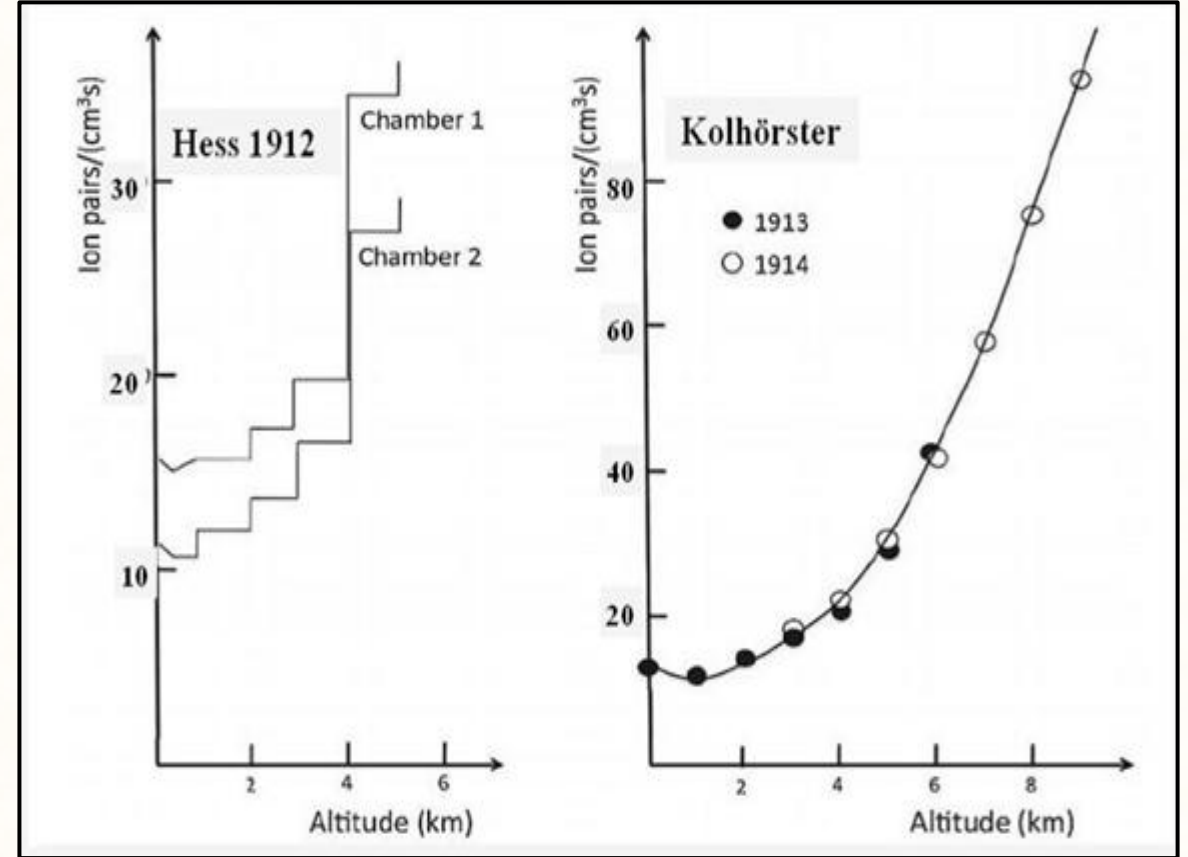
- The common belief was that the discharges were mainly coming from gamma rays. New detection techniques and studies on radiation sources found that they couldn't explain the observations
- A **change of radiation level with latitude** was also found... different radiation levels in different parts of the Earth's surface?
- A few experiments showed that the radiation levels were not very different when the detectors were placed on boats
- Others when up in towers or on mountains, finding inconclusive results



Where do cosmic rays come from?

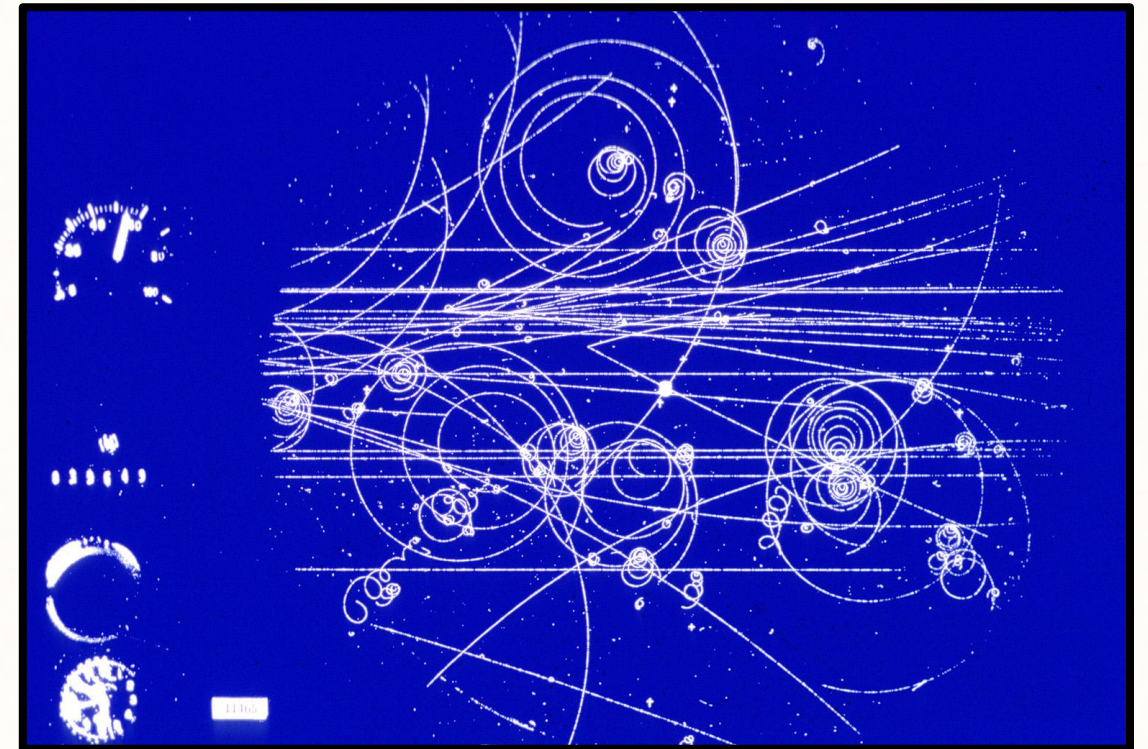
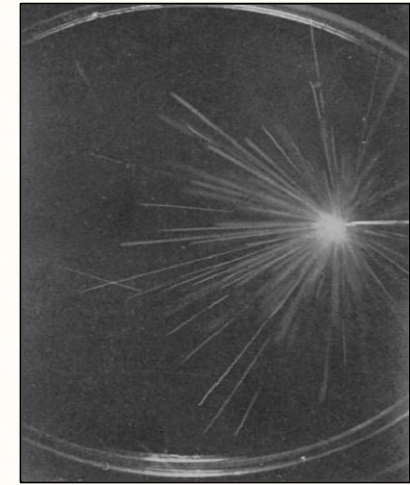
- **Experiments on balloons resulted key!**
- An increase of radiation level with altitude was clear
- When going inside a lake the radiation decreased too (Domenico Pacini)

“When I was able to demonstrate that the ionization was reduced with increasing height, I concluded that this ionization might be attributed to the penetration of the earth’s atmosphere from outer space by hitherto unknown radiation of exceptionally high penetrating capacity” – Victor Hess, 1936



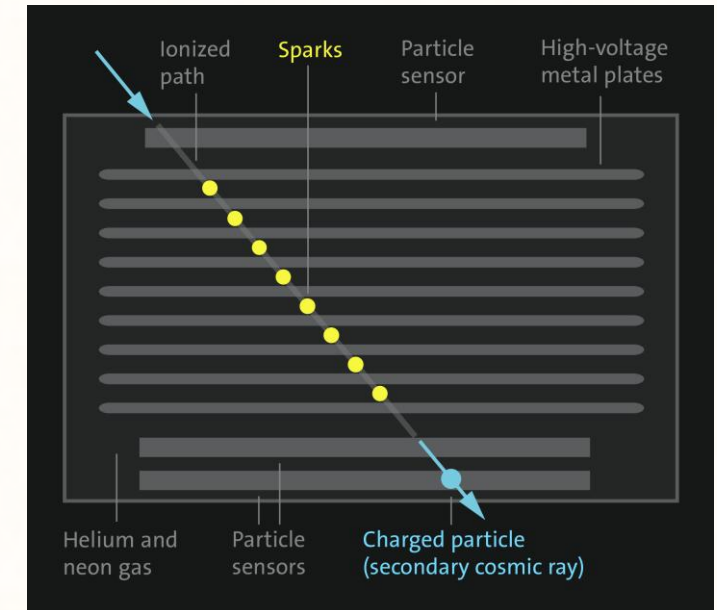
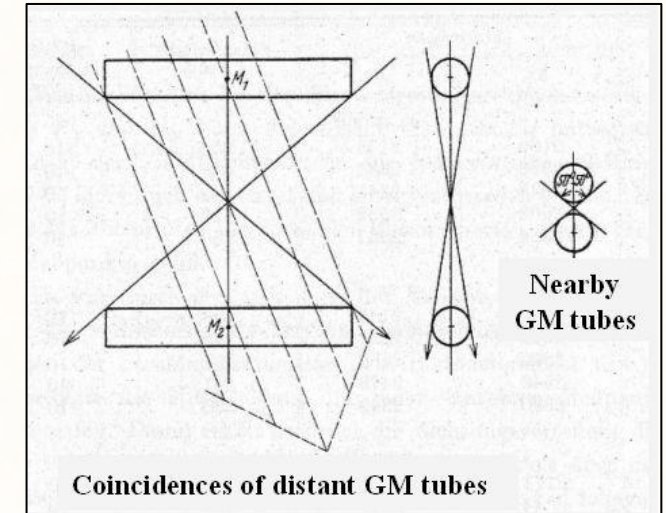
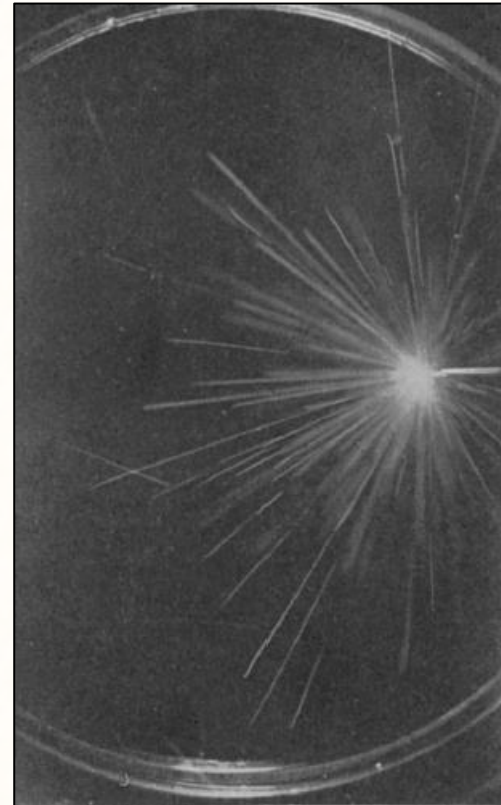
What are cosmic rays?

- Then, better instruments (especially the **cloud chamber**) allowed us to really see the particle tracks and identify the particles passing through the detector. We could see **cosmic ray tracks**!
- Coincidence detectors allowed us to measure the energy and direction of the particles.
- **Cosmic-ray showers** were observed!
- And particle physics was born:
 - Positrons (1912)
 - Muons (1936)
 - Pions and Kaons (1947)



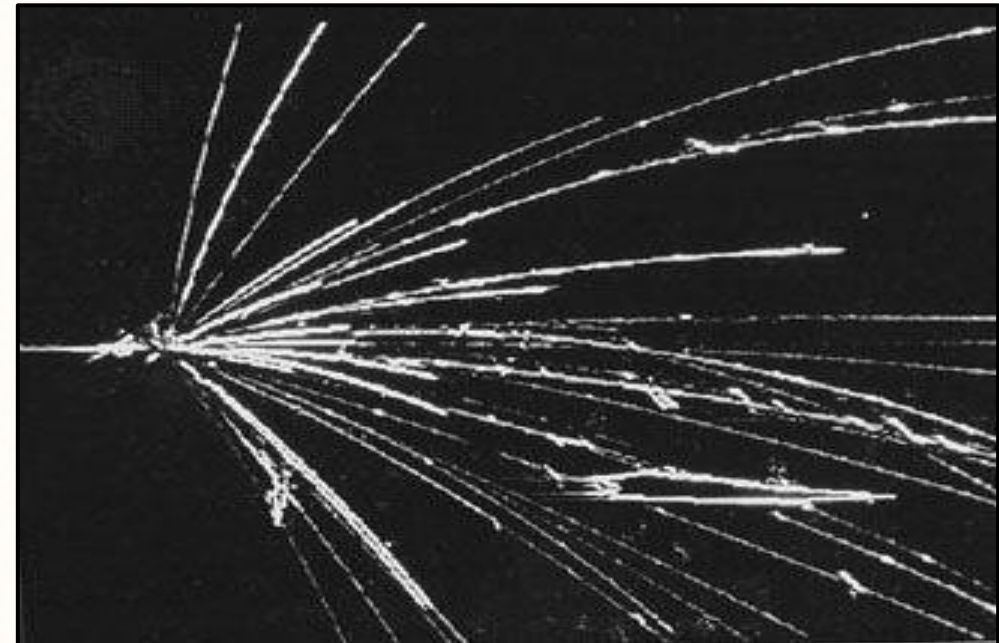
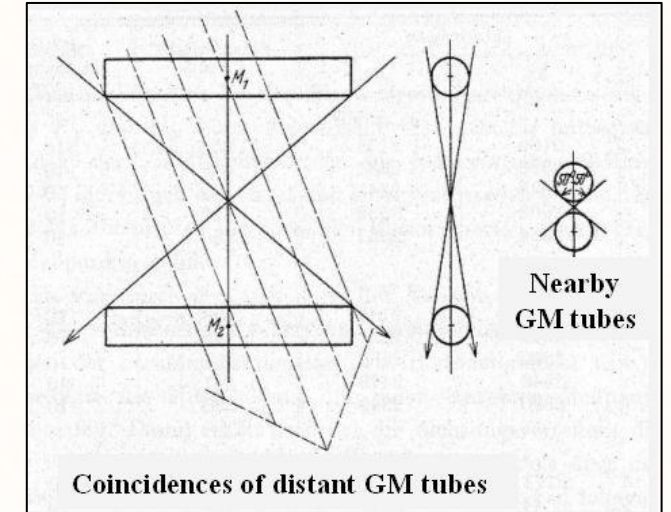
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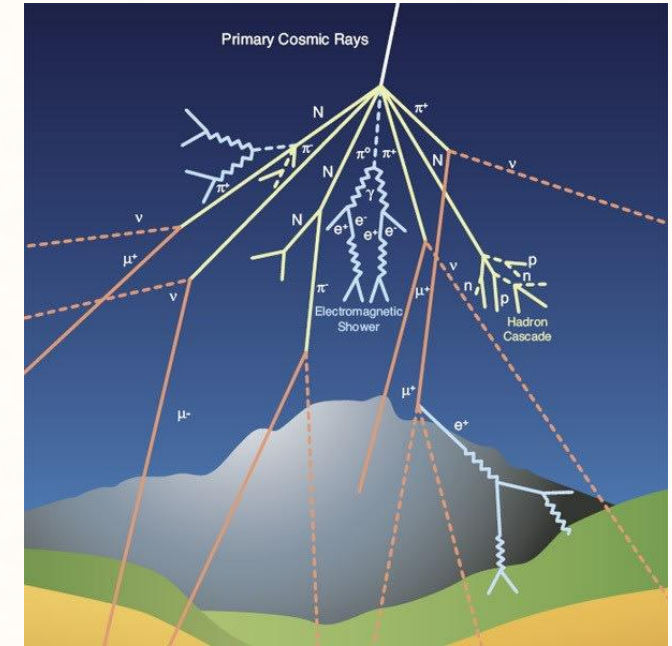
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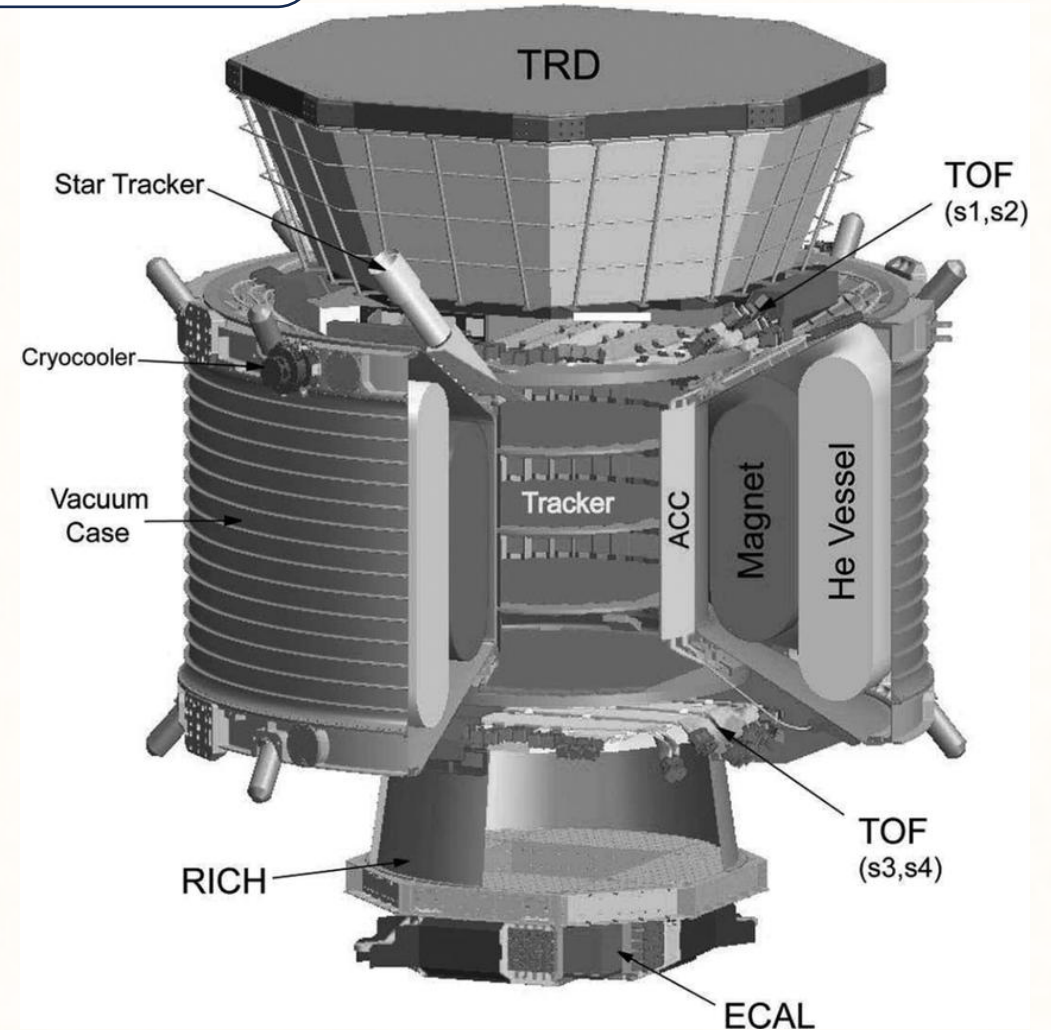
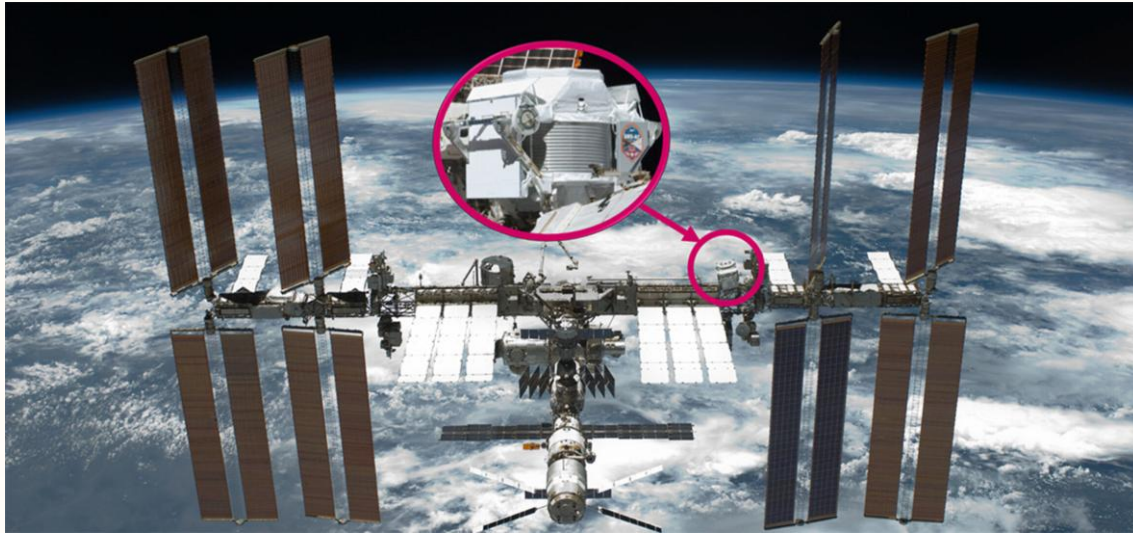
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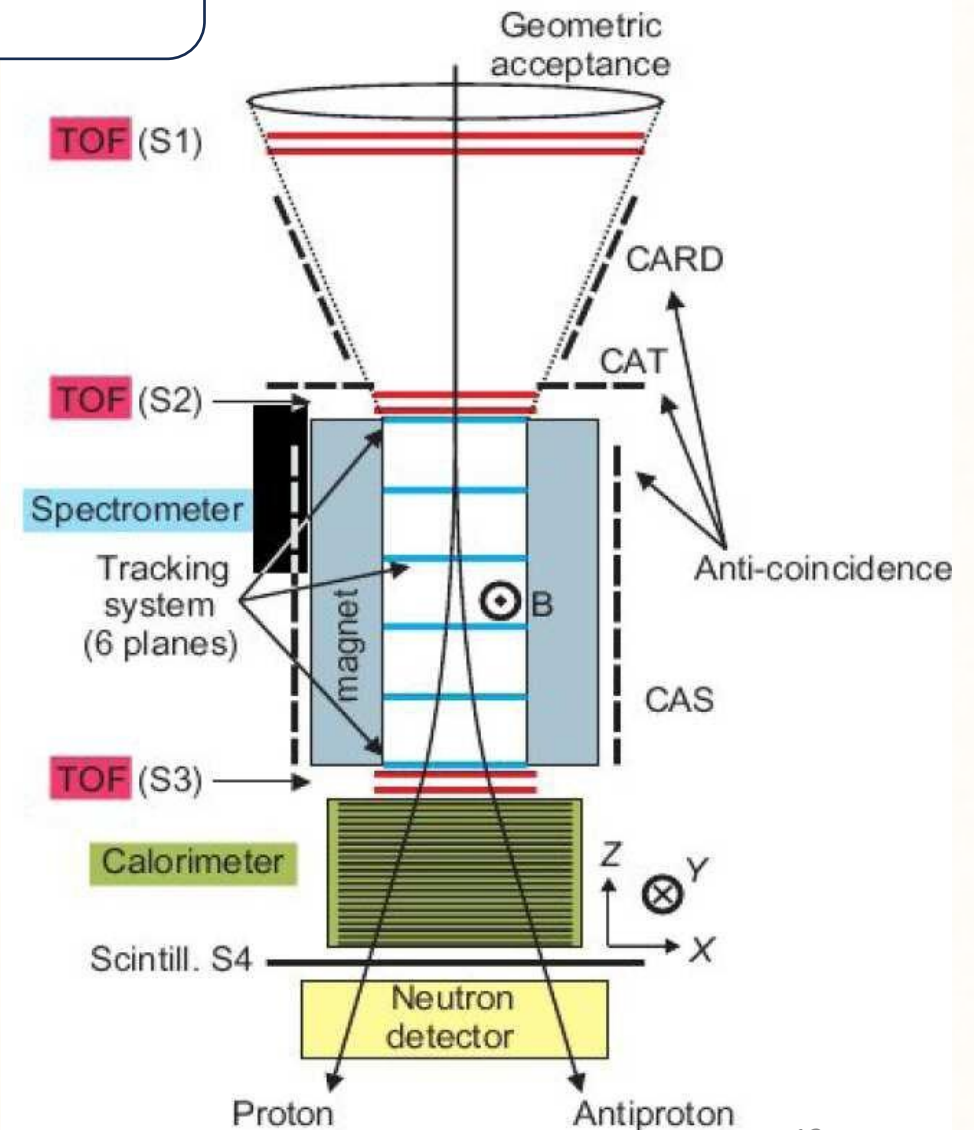
Detectors today - Spectrometers

The **AMS-02 (Alpha magnetic spectrometer)**, on the International Space Station (ISS)



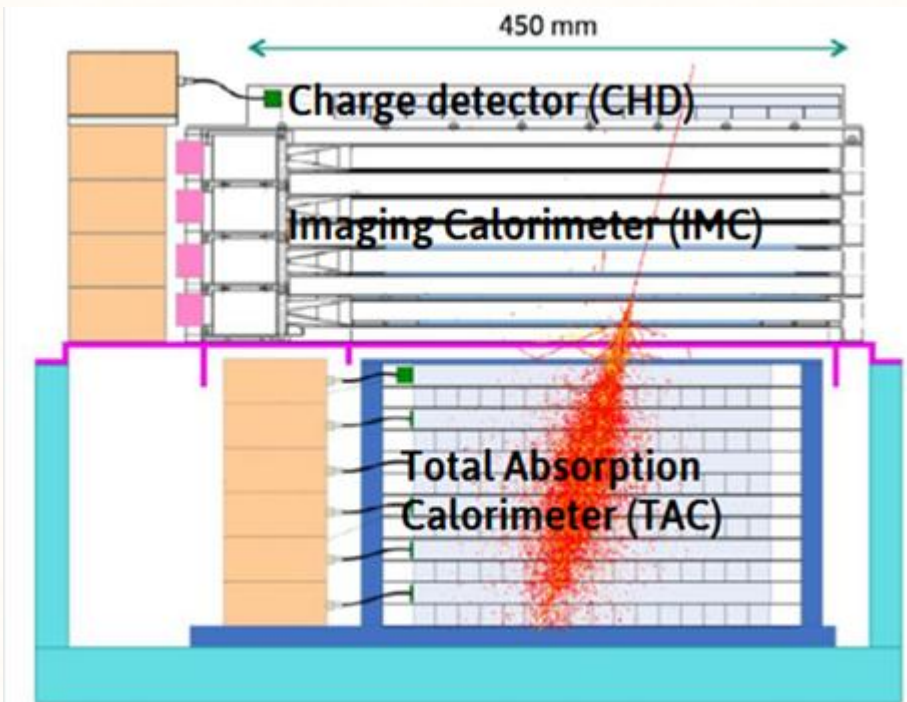
Detectors today - Spectrometers

PAMELA, in a satellite orbiting the Earth

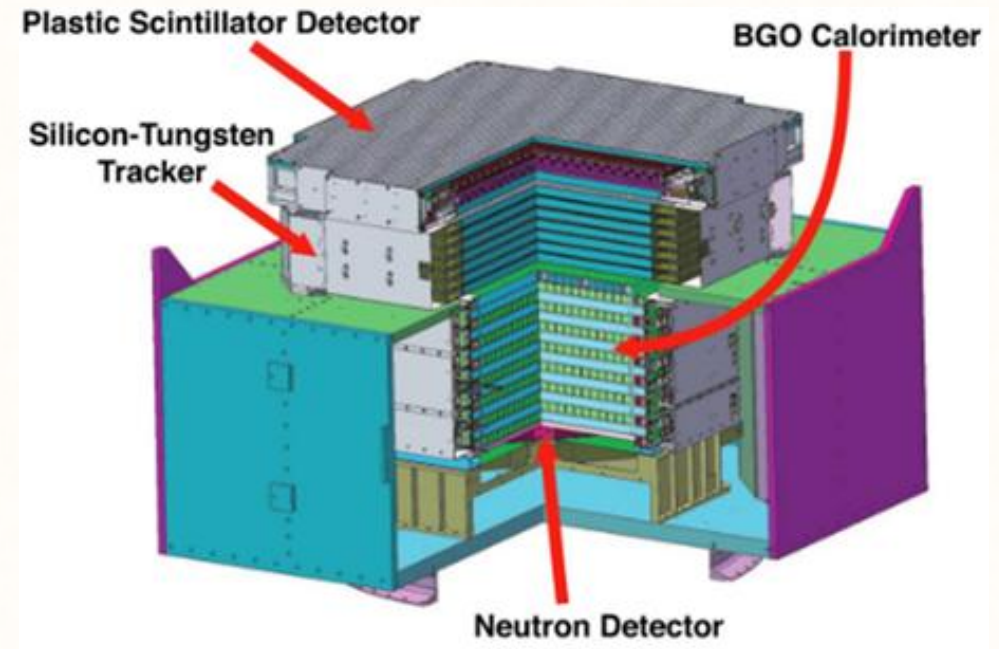


Detectors today - Calorimeters

CALET (CALorimeter Electron Telescope) on the ISS



DAMPE (DARK Matter Particle Explorer) - Satellite

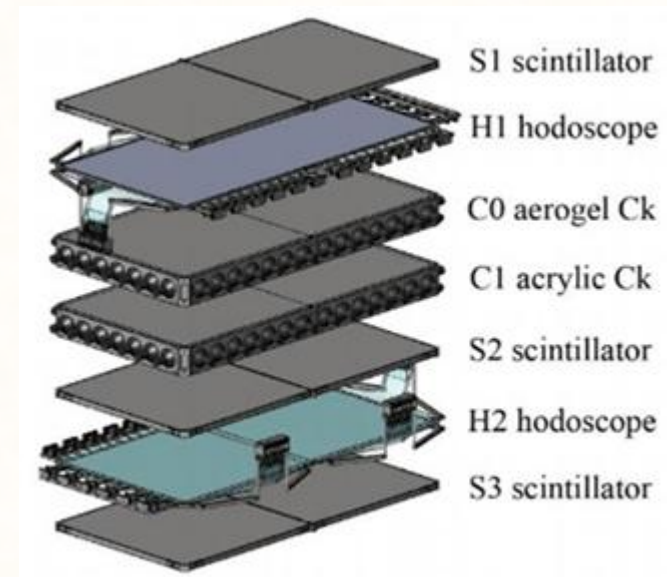


Detectors today - Balloons

CREAM (Cosmic Ray
Energetics and Mass)

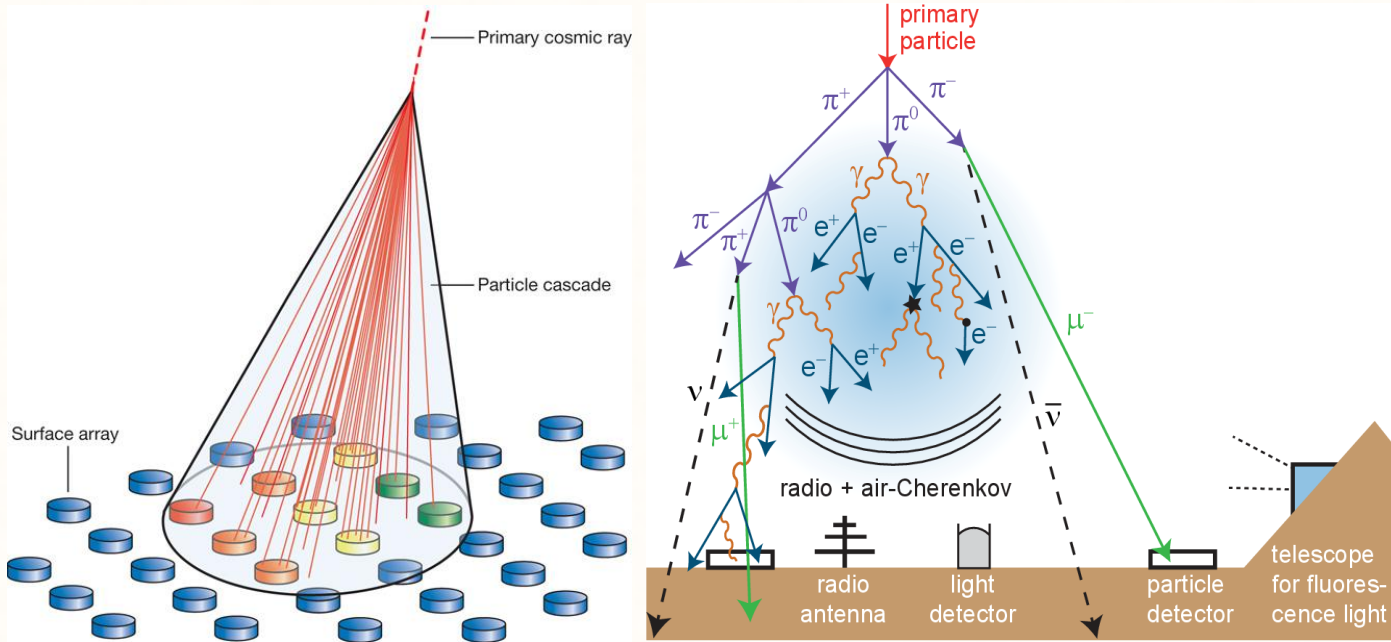


Super-TIGER (Super Trans-Iron
Galactic Element Recorder)



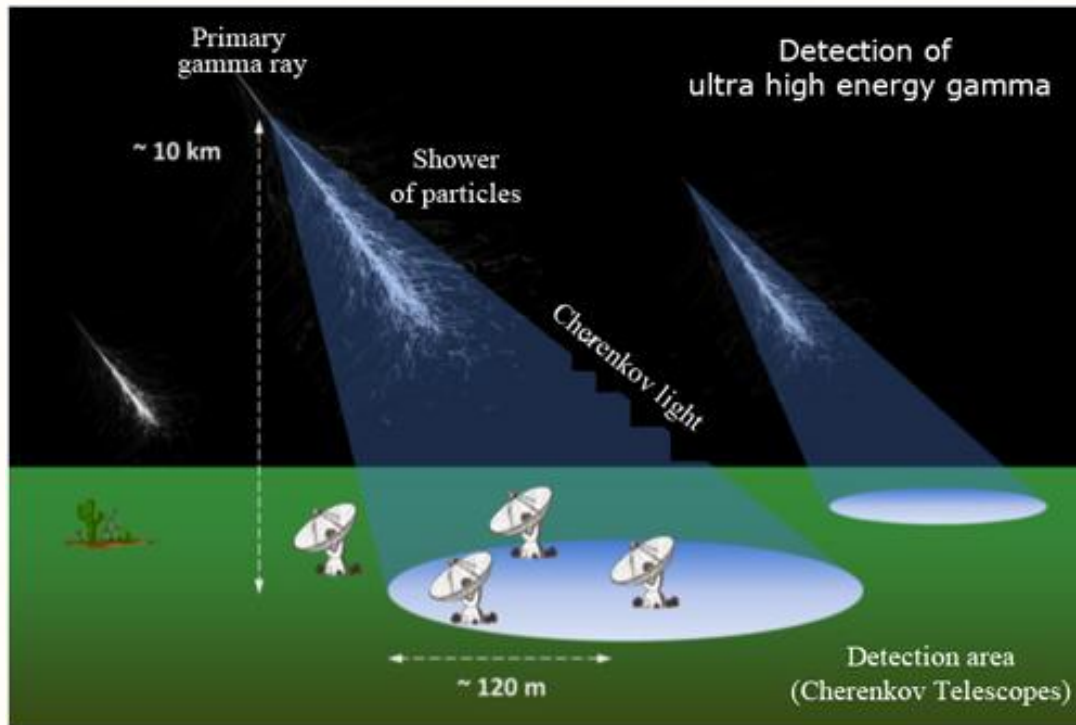
Detectors today – shower detectors

Pierre Auger Observatory or the **Telescope Array**, measure ultra-high energy CRs



Detectors today – shower detectors

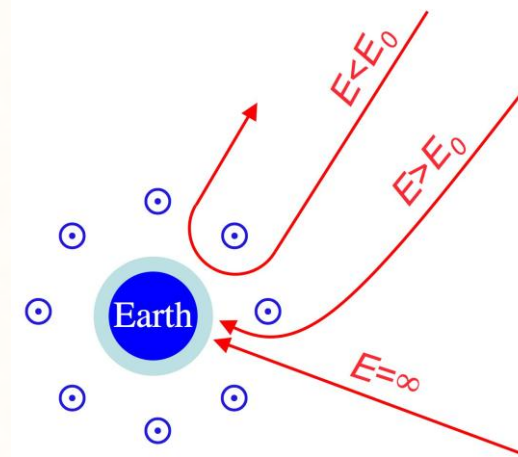
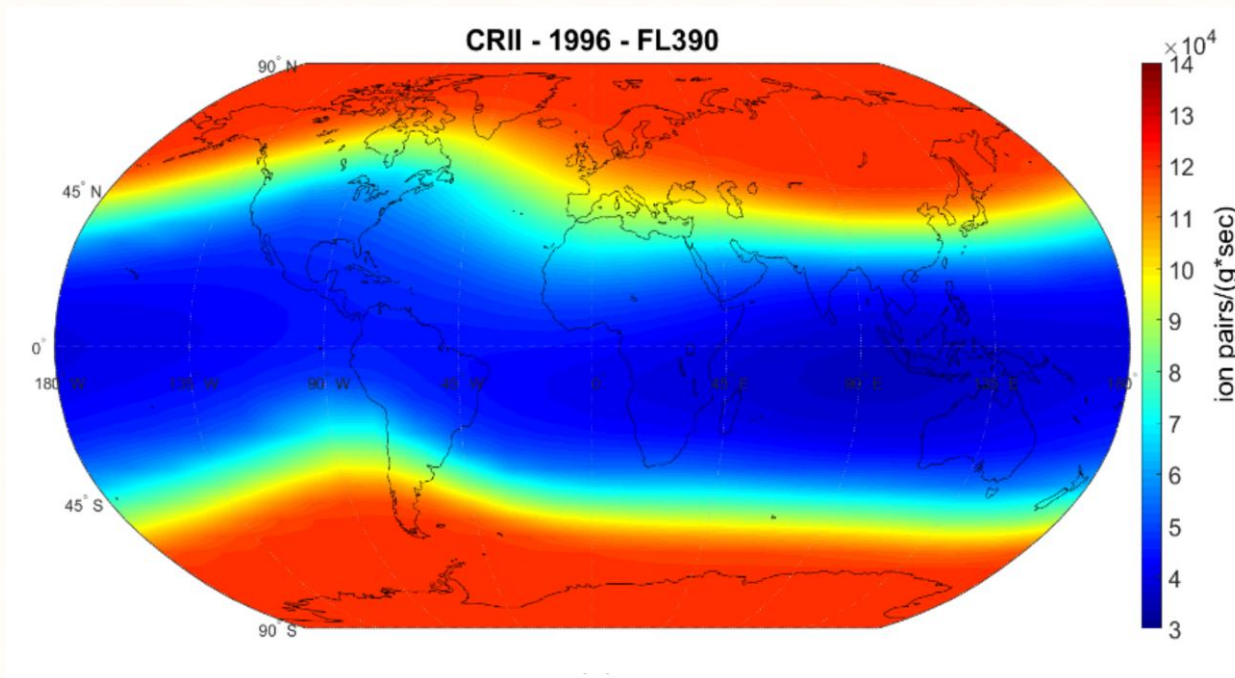
HESS, VERITAS, MAGIC, ... measure the Cherenkov light emitted in electromagnetic particle showers



What we observe today

Such a variety of detectors and detection techniques allows us to characterize the flux and composition of cosmic rays at Earth

Does the flux of CRs depend on their arrival direction?
We observe variations in the CR flux with latitude and longitude. These are due to the geomagnetic field, which deflects the trajectories of particles and “shields” the equator from low-energy CRs (geomagnetic cutoff). Accounting for this, only tiny variations are found.



What we observe today

Other anisotropies in CR arrival directions?

GeV cosmic rays generally arrive from all directions, but display a small, persistent dipole anisotropy caused by the solar system's 200 km/s motion through the galaxy. This Compton-Getting effect produces a slight excess of cosmic rays in the direction of motion, with dipole amplitude increasing towards 200 GeV.

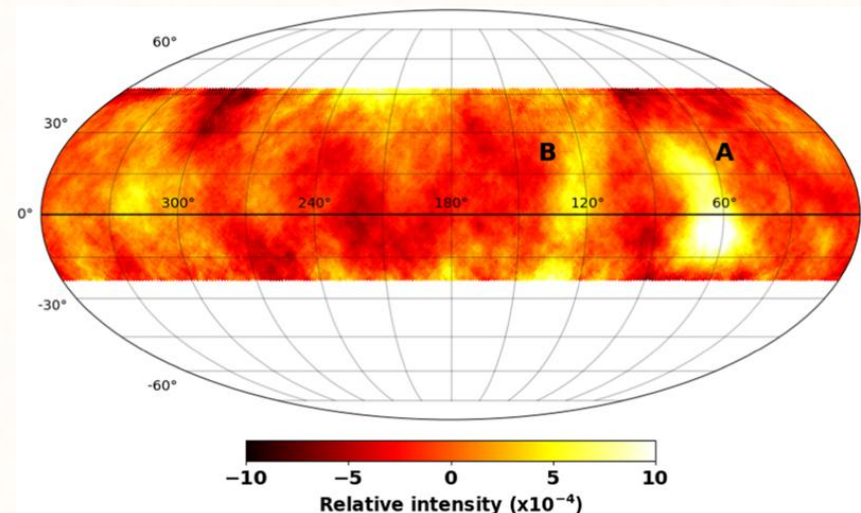
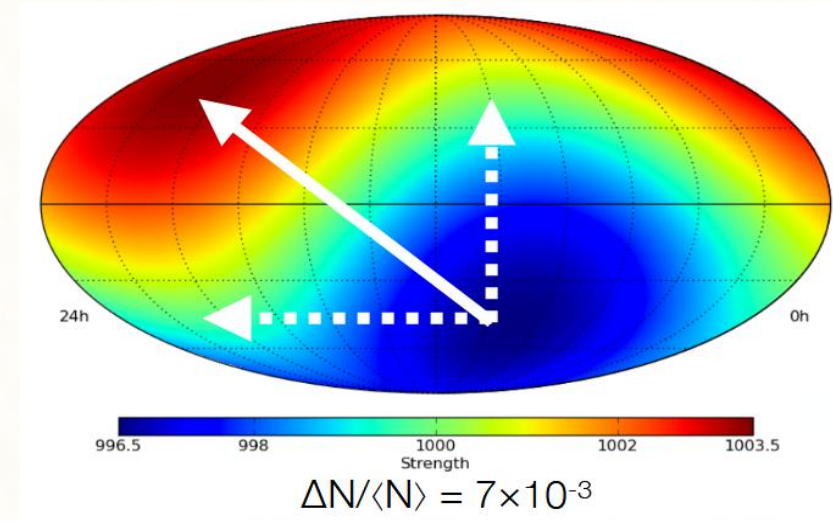
After accounting for this effect and the geomagnetic field, the level of anisotropy observed in cosmic rays is really low: **They come evenly from everywhere**

$\delta < 1.44\%$ for positrons ($16 < E/\text{GeV} < 500$)

$\delta < 0.39\%$ for electrons ($16 < E/\text{GeV} < 500$)

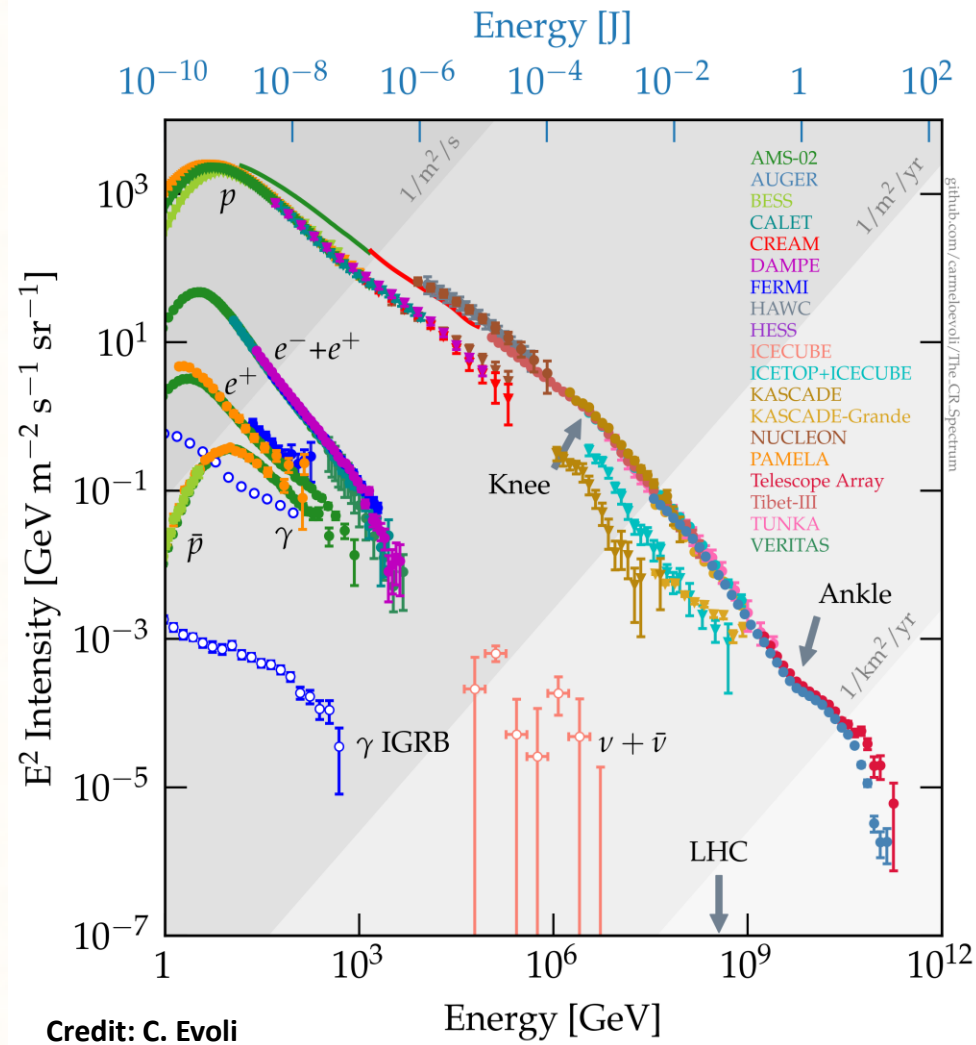
$\delta < 0.30\%$ for protons ($R > 200 \text{ GV}$)

Such a variety of detectors and detection techniques allows us to characterize the flux and composition of cosmic rays at Earth

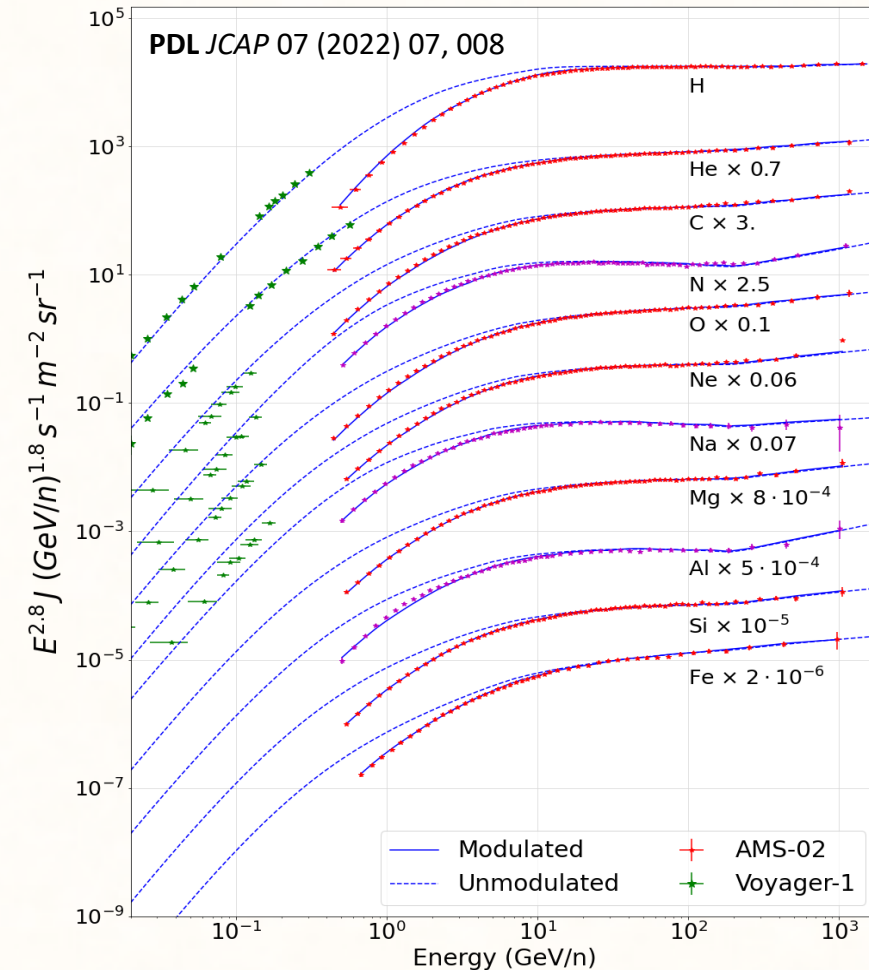


What we observe today

Such a variety of detectors and detection techniques allows us to characterize the flux and composition of cosmic rays at Earth



CRs follow a nearly common power law behaviour



Cosmic rays: *Those particles coming from the outer space that constitute a constant and steady, nearly isotropic (to the Earth perspective), flow*

The CR flux is given as the number of cosmic rays arriving per energy interval (dE), solid angle ($d\Omega$), per unit time and per unit area (A). When we refer to their spectrum, we mean the behaviour of this flux with energy

$$\Phi(E) \equiv \frac{dN}{A \cdot T \cdot d\Omega \cdot dE} \frac{\text{particles}}{\text{cm}^2 \text{sr s GeV}} = K \left(\frac{E}{1 \text{ GeV}} \right)^{-\alpha} \quad K = 3.01; \quad \alpha = 2.68$$

Depending on the specific problem, we usually report this spectrum as function of momentum, kinetic energy, kinetic energy per nucleon (because in certain particle interactions important for cosmic rays, this quantity is conserved) or rigidity (because CRs with the same rigidity behave in the same way under electromagnetic forces)

$$E^2 = p^2 c^2 + E_0^2$$

$$E = T + E_0$$

Energies per nucleon:

$$A^2(T + E_0)^2 = p^2 c^2 + A^2 E_0^2$$

$$P = pc/(Ze) = (A/Ze) \sqrt{T(T + 2E_0)} = (A/Ze) \beta(T + E_0)$$

The origin of Cosmic Rays

- **Some supported that CRs came from the Big Bang**

- Their composition does not coincide with that expected from big bang nucleosynthesis (much closer to the composition of the interstellar medium)
- We would expect a kind of thermal spectrum
- It leads to other contradictions as well

- **Is the Sun a source of CRs?**

- Temporal coincidence with solar flare activity and particle fluxes at low energies (sub-GeV).
- Also, a small time-dependence is found in CRs

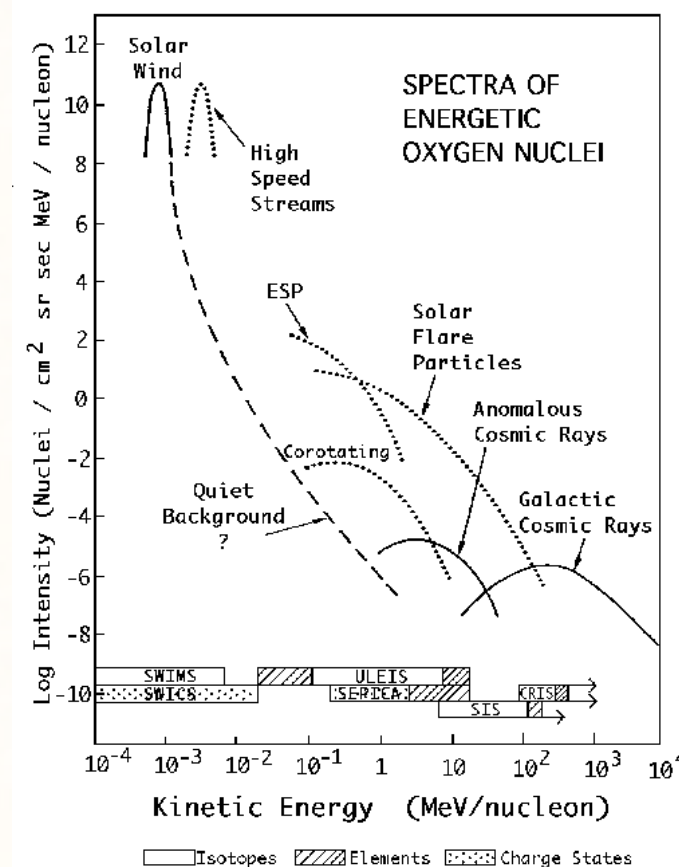
- **A Galactic origin seems plausible!**

- Bright astrophysical sources are a natural candidate
- Zwicky pointed to **supernovae** as a promising candidate

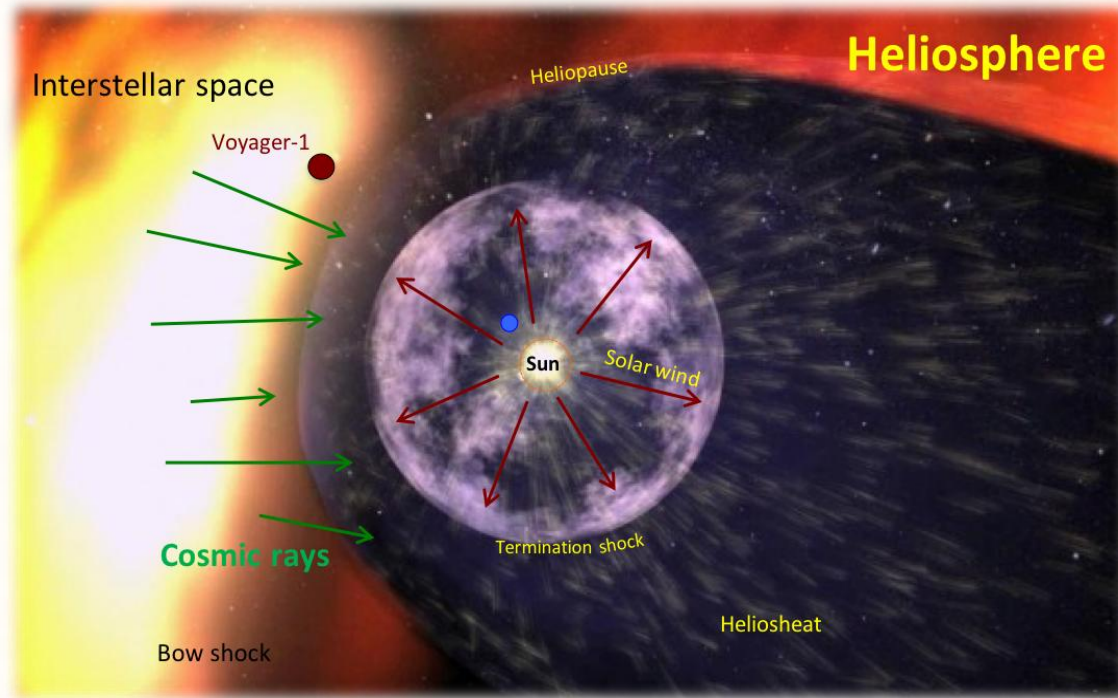
$$\rho_{\text{CR}} \equiv \frac{1}{c} \int_{E_0}^{\infty} E \frac{d^2\varphi}{dE d\Omega} dE d\Omega \simeq 1 \text{ eV/cm}^3$$

$$\rho_{\gamma\text{CMB}} \sim 0.3 \text{ eV/cm}^{-3}$$

$$\rho_B = \frac{1}{8\pi} B^2 \text{ erg/cm}^3 \simeq 1 \text{ eV/cm}^3$$



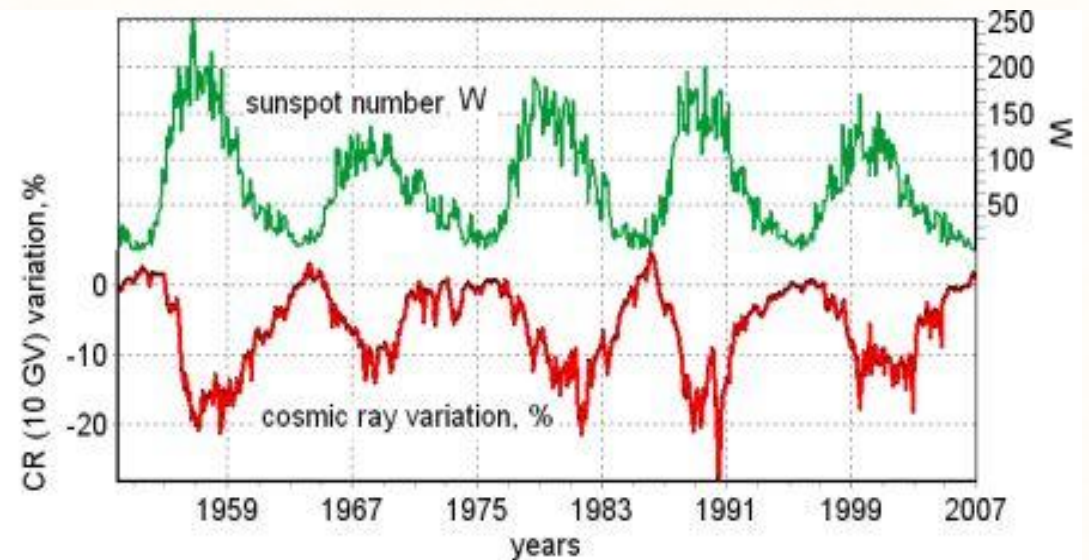
Effect of the Heliosphere – Solar modulation



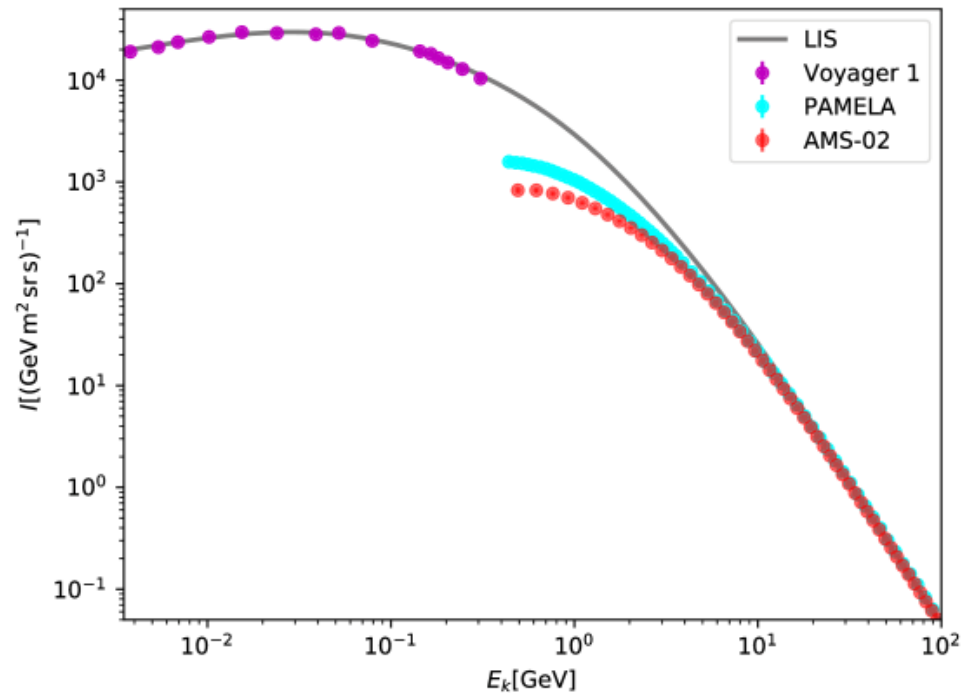
Low-energy cosmic rays are more affected by the shielding caused by the outward solar wind
Their interaction produces a time dependence perfectly anticorrelated with the Sun's activity

CRs experience a “firewall” when they enter the heliosphere from interstellar space

It significantly affects the propagation of low-energy CRs (below $E \sim 10$ GeV/n)



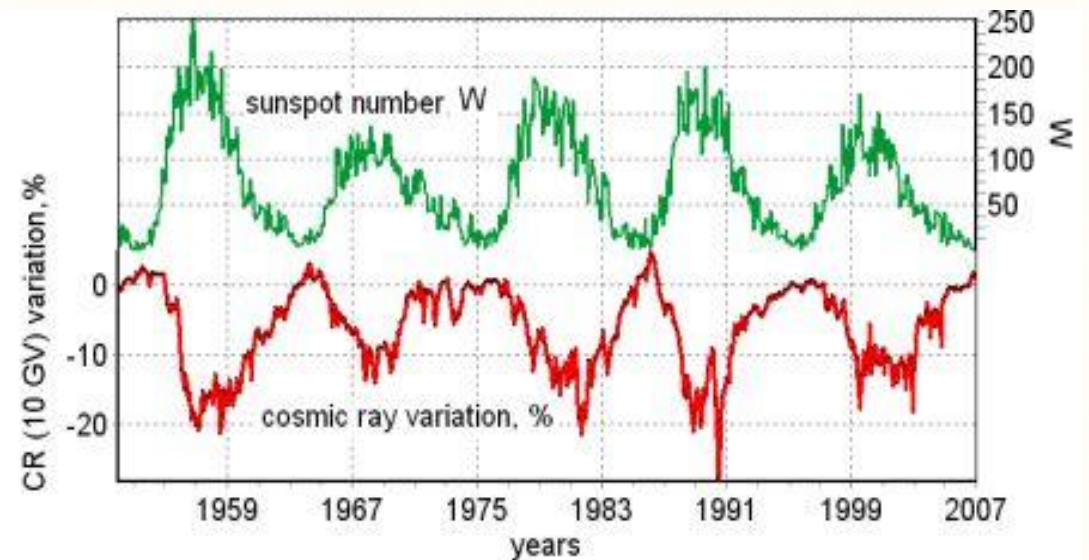
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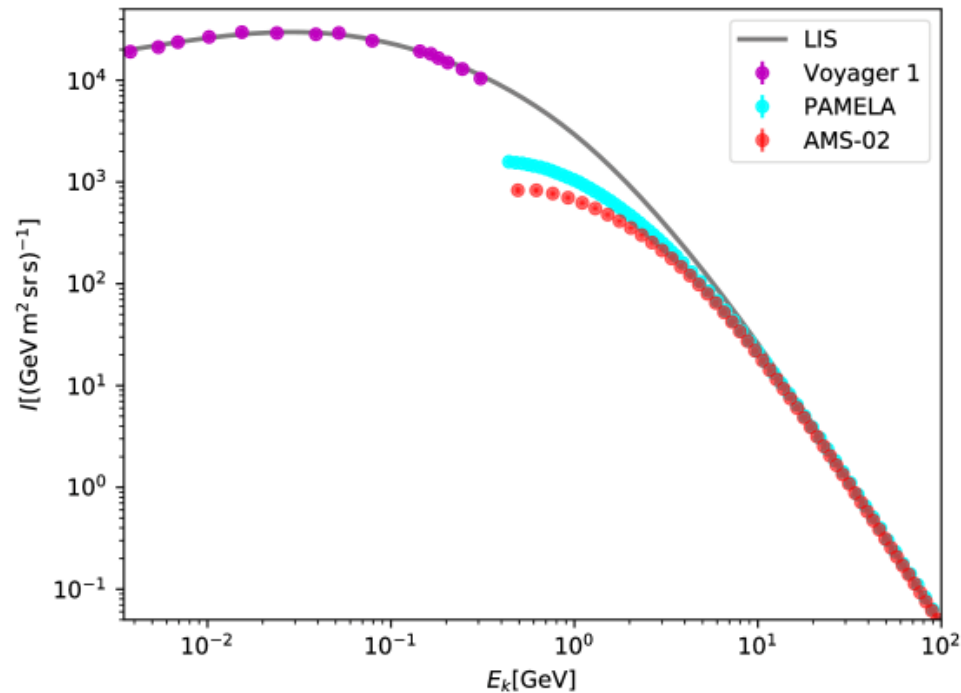
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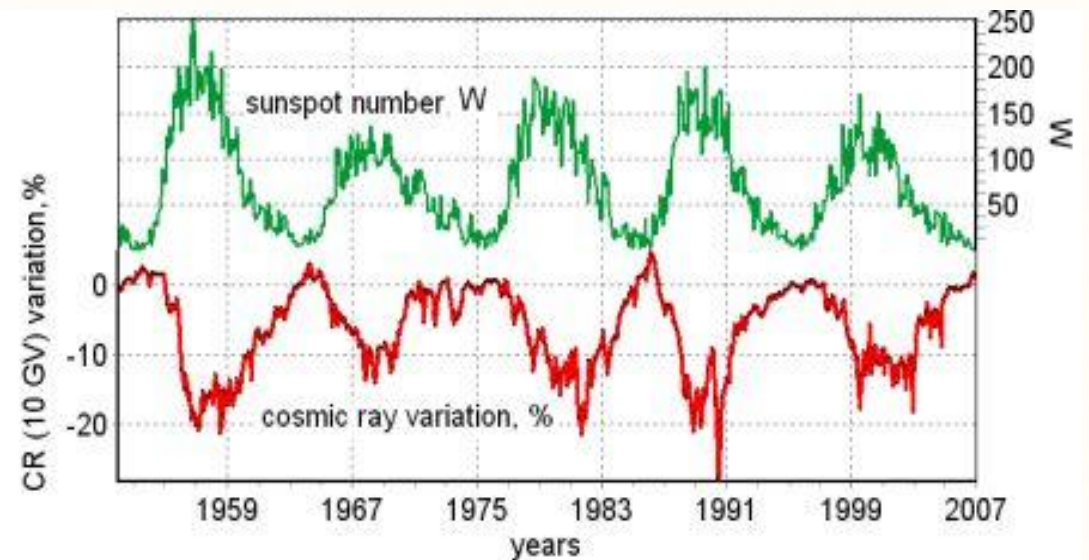


Effect of the Heliosphere – Solar modulation



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It significantly affects the propagation of low-energy CRs (below $E \sim 10$ GeV/n)



$$J_{\oplus}(E) = \frac{E^2 - m^2}{(E + |Z|e\phi)^2 - m^2} J_{\text{LIS}}(E + |Z|e\phi)$$

$$T_{\text{LIS}} = T_{\oplus} + |Z|e\phi$$

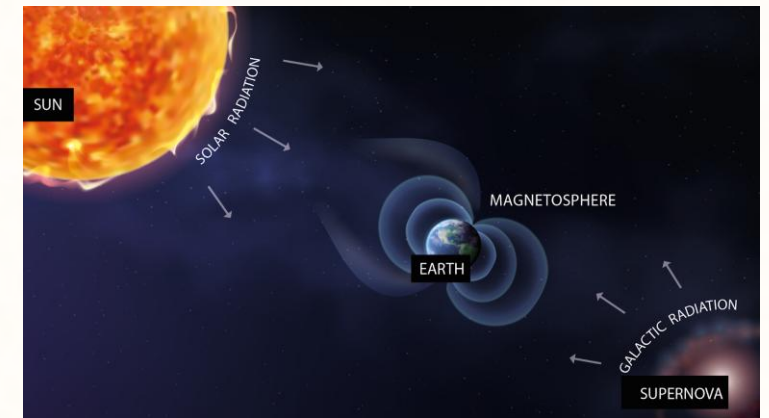
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$$\rho_{\gamma\text{CMB}} \sim 0.3 \text{ eV/cm}^{-3}$$

$$\rho_B = \frac{1}{8\pi} B^2 \text{ erg/cm}^3 \simeq 1 \text{ eV/cm}^3$$



$$\rho_{\text{CR}} \times \mathcal{V}_G = 8 \cdot 10^{54} \text{ erg.}$$

$$P_{\text{CR}} \simeq \frac{\rho_{\text{CR}} \times \mathcal{V}_G}{\tau_{\text{esc}}} = \frac{8 \cdot 10^{54}}{3 \cdot 10^{14}} = 3 \times 10^{40} \text{ erg/s}$$

$$P_{\text{SN}} \simeq \eta \times f_{\text{SN}} \times 10^{51} = \eta \times 10^{42} \text{ erg/s}$$

How to accelerate Cosmic Rays?

Enrico Fermi showed that the interaction of charged particles with magnetic irregularities in a plasma medium (eddies) can lead to acceleration (or deceleration) of cosmic rays.

Key finding: Statistical gain of energy by particles, naturally results in a power-law distribution.

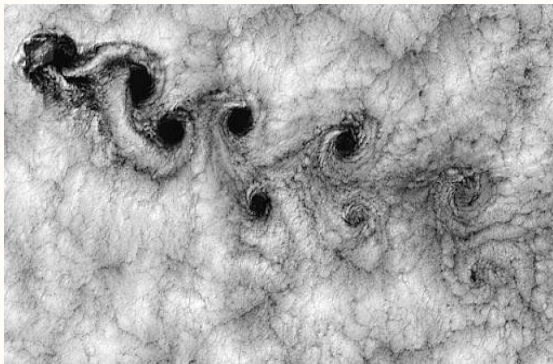
Fermi 2nd order mechanism (the intuition). Imagine a series of encounters with eddies:

Sometimes a CR hits an eddy moving **toward it** → it gains energy

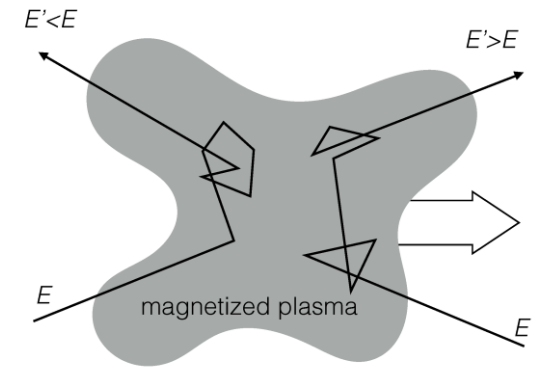
Sometimes it hits one moving **away** → it loses energy

But on average, it gains a little more energy than it loses

$$\left\langle \frac{\Delta E}{E} \right\rangle = \frac{4}{3} \left(\frac{V}{c} \right)^2 > 0$$



Head-on encounters are more probable because collision probability scales with relative speed, and relative speed is larger when particle and eddy move toward each other.

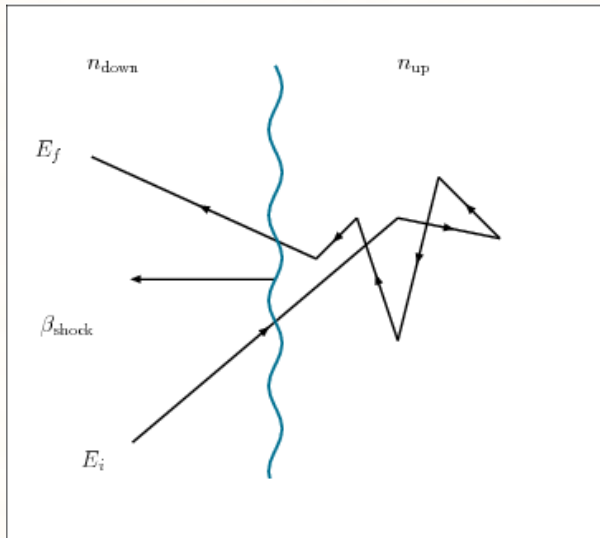


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Fermi 1st order mechanism (the intuition). Eddies always move in the same direction:
Shock fronts create a discontinuity, separating a highly turbulent medium from a quieter plasma



$$\left\langle \frac{\Delta E}{E} \right\rangle \approx \frac{4}{3} \frac{u_{\text{shock}}}{c}$$

It provides an efficient way to accelerate CRs up to high energies and have a power-law spectra

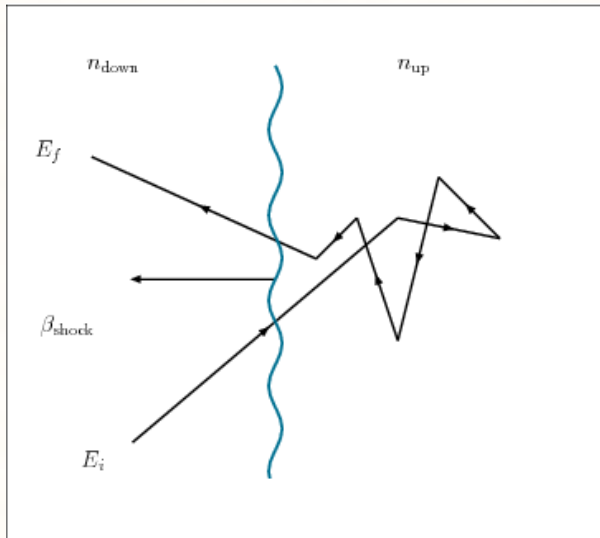


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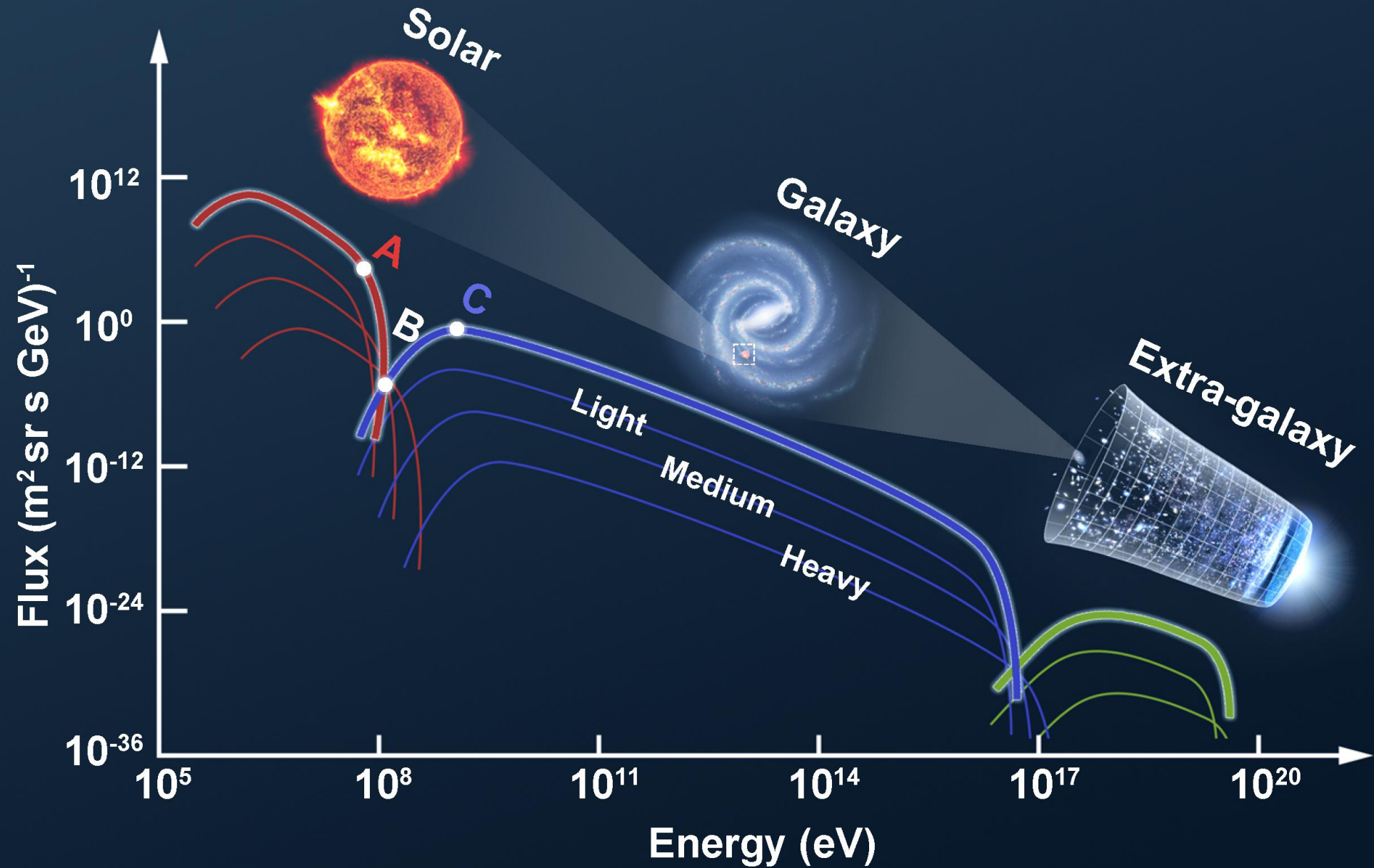
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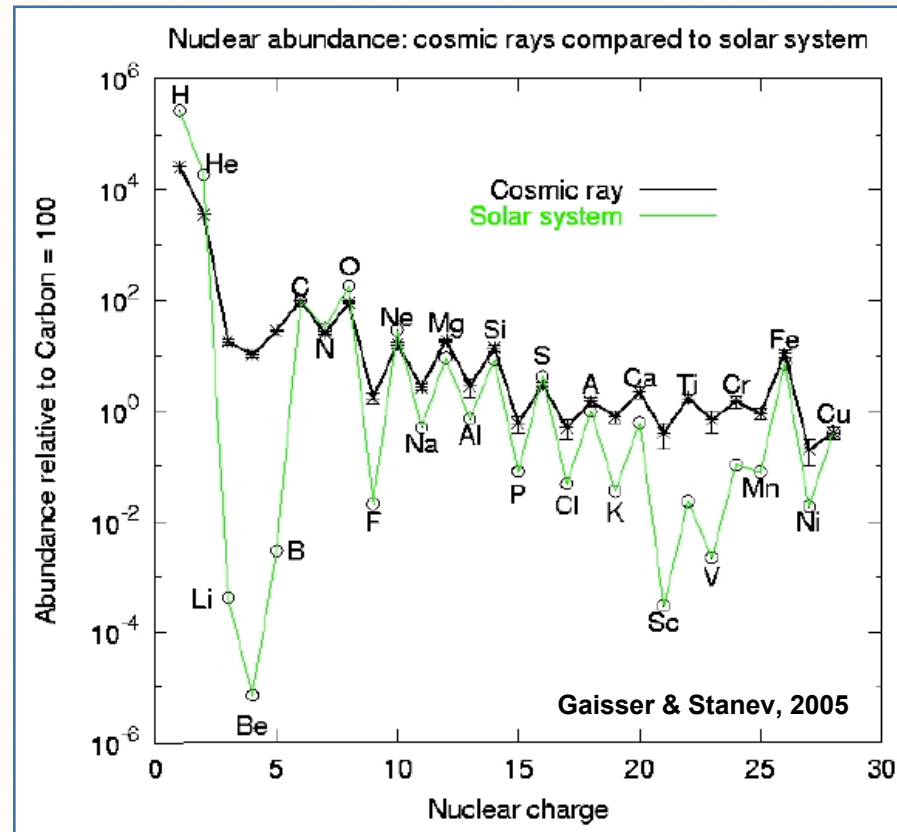




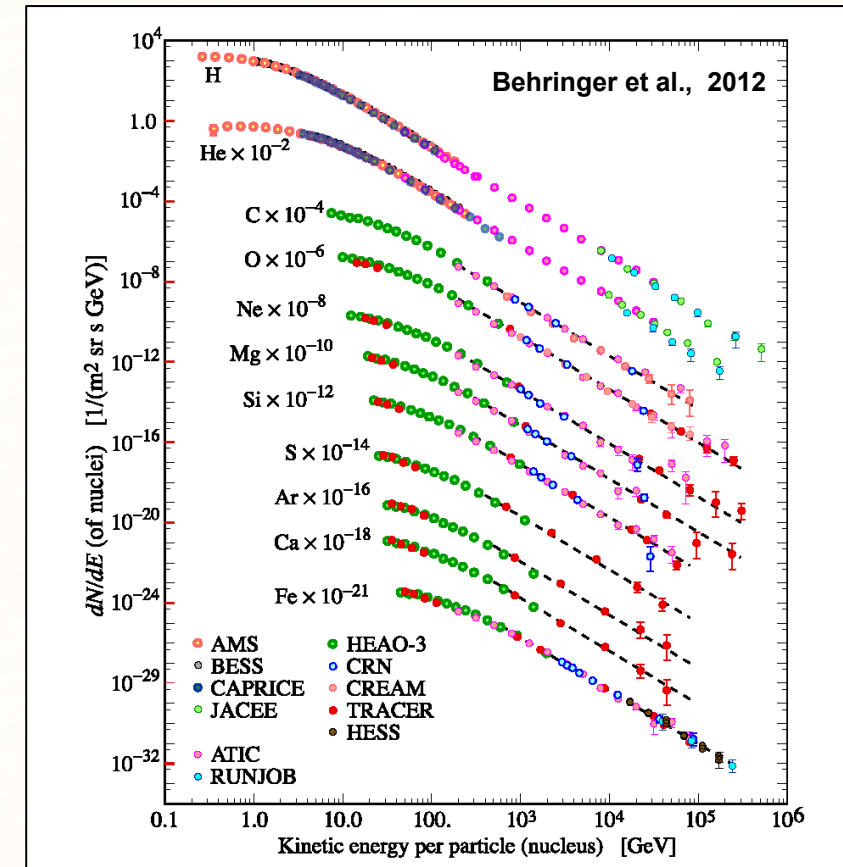
However, there are at least three main problems:

- 1) The spectral index found is always much larger to what shocks in SNRs can explain
- 2) But the spectral index for B, Be, Li and even F is even larger
- 3) Relative composition of cosmic rays, compared to that of the interstellar medium

Unexpected abundance of secondary nuclei



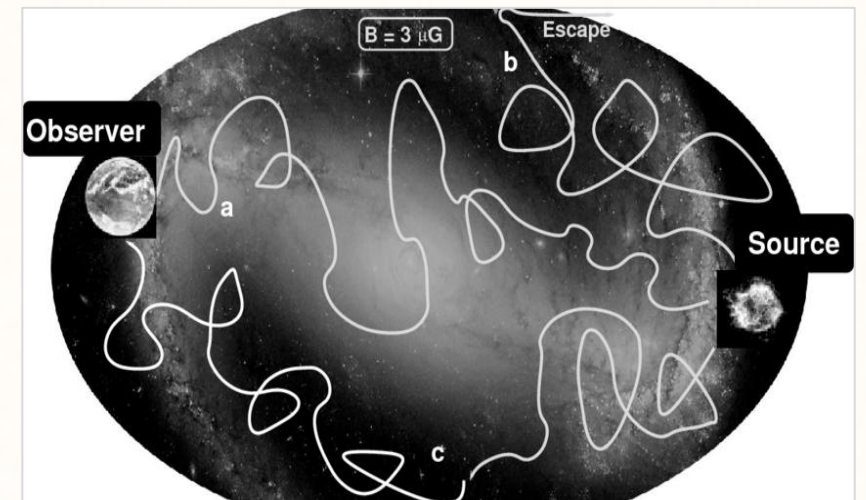
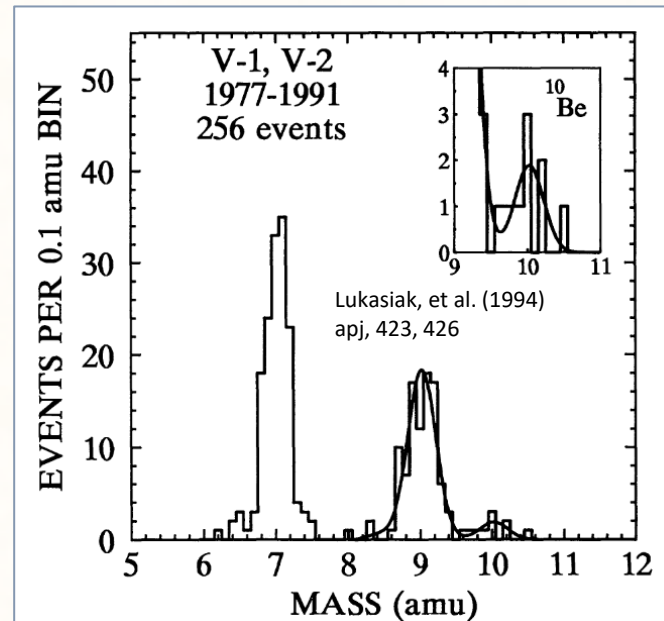
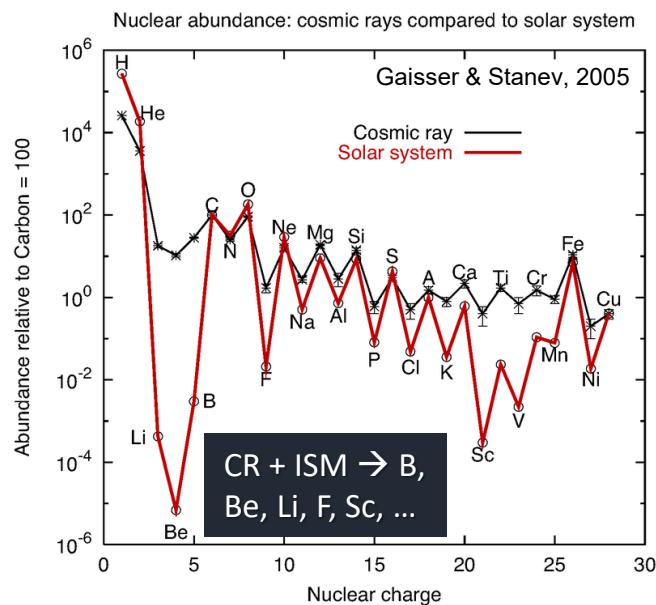
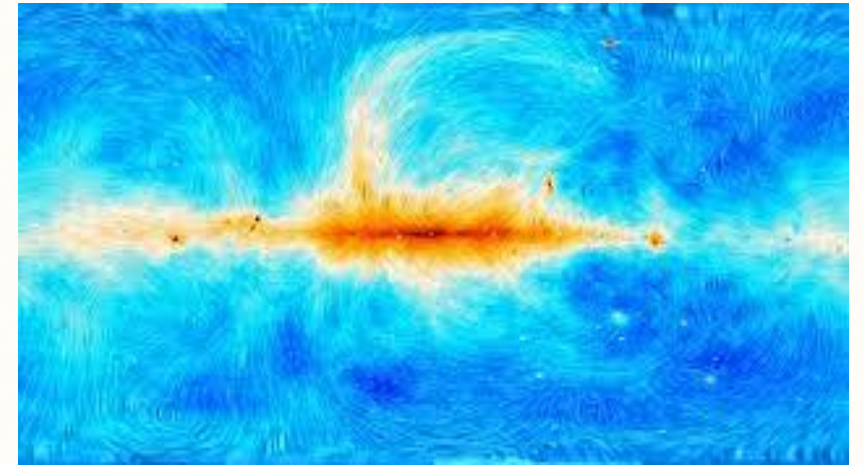
Steeper power laws than predicted



A few key observations show evidence that CRs propagate in the Galaxy for millions of years

Charged particles are accelerated by astrophysical sources and propagate throughout the Galaxy during millions of years, due to scattering with plasma waves.

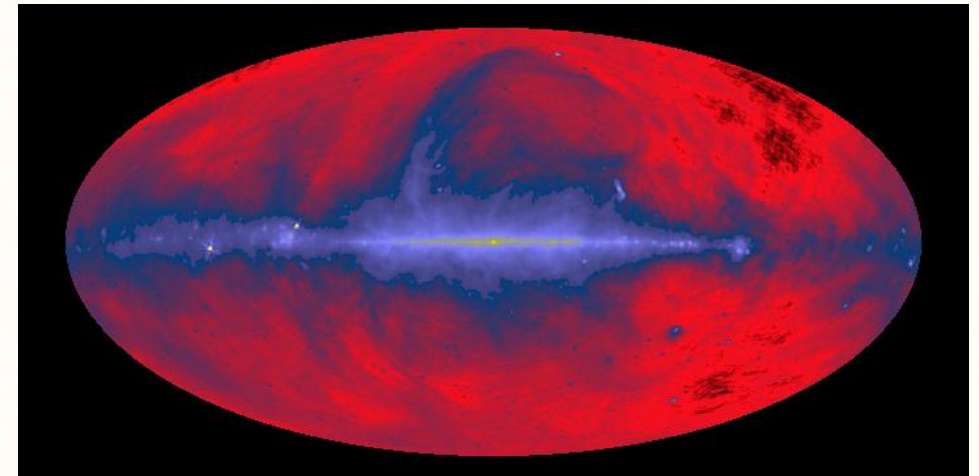
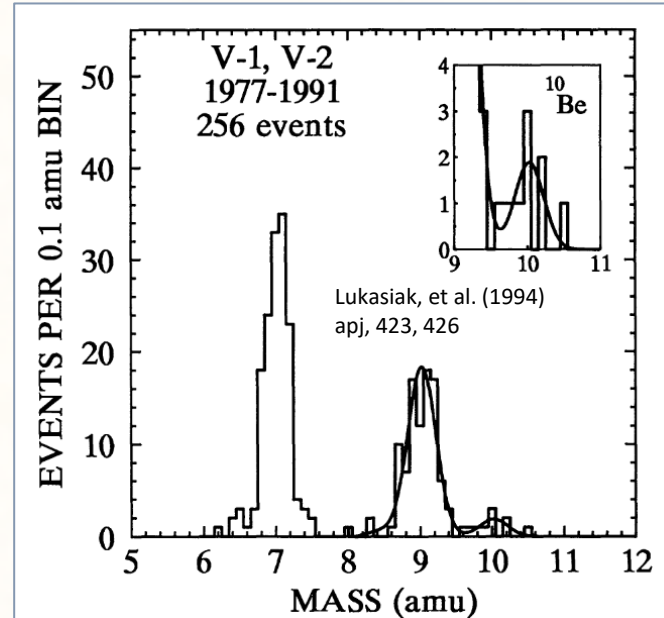
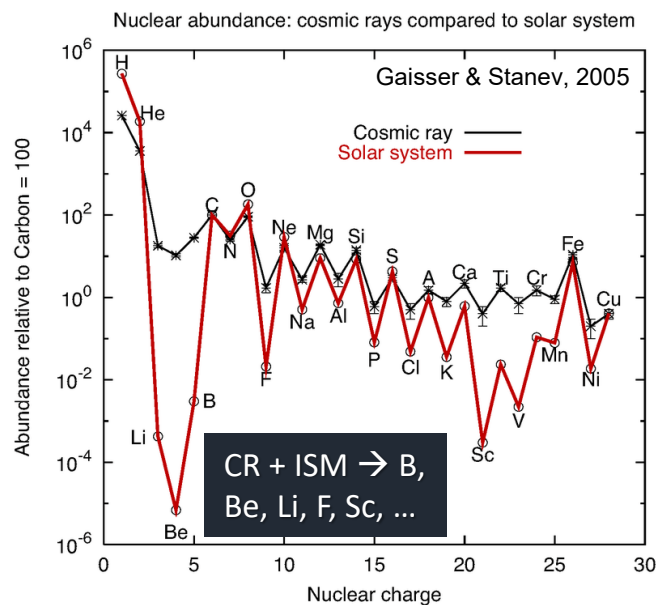
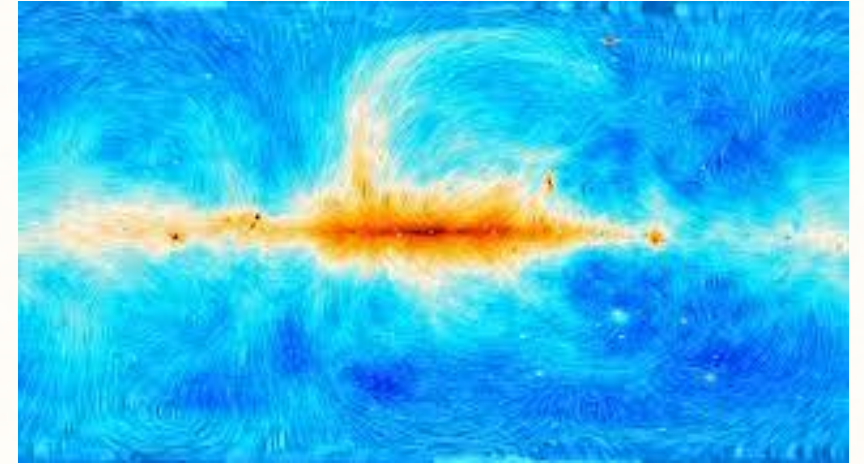
Cosmic-ray transport must be governed by their interactions with magnetic fluctuations!



A few key observations show evidence that CRs propagate in the Galaxy for millions of years

Charged particles are accelerated by astrophysical sources and propagate throughout the Galaxy during millions of years, due to scattering with plasma waves.

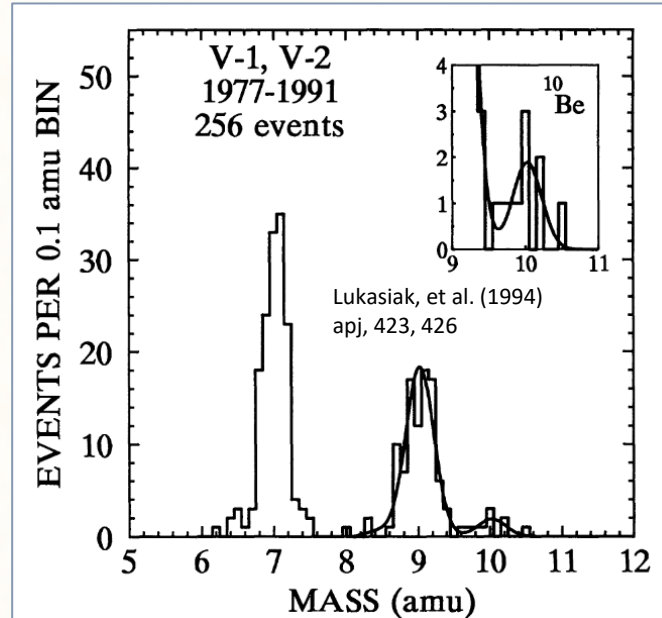
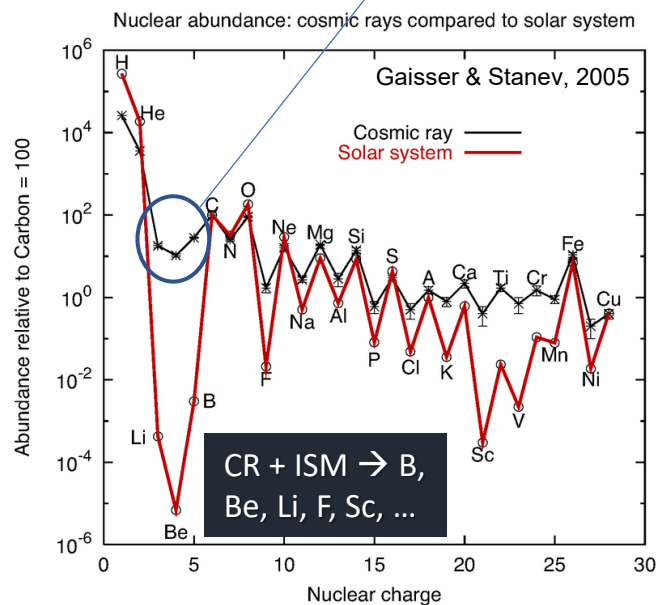
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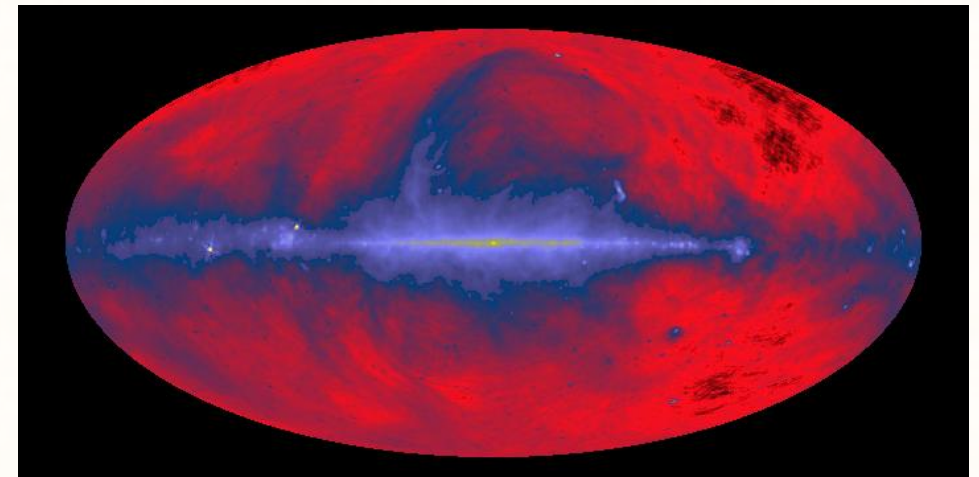
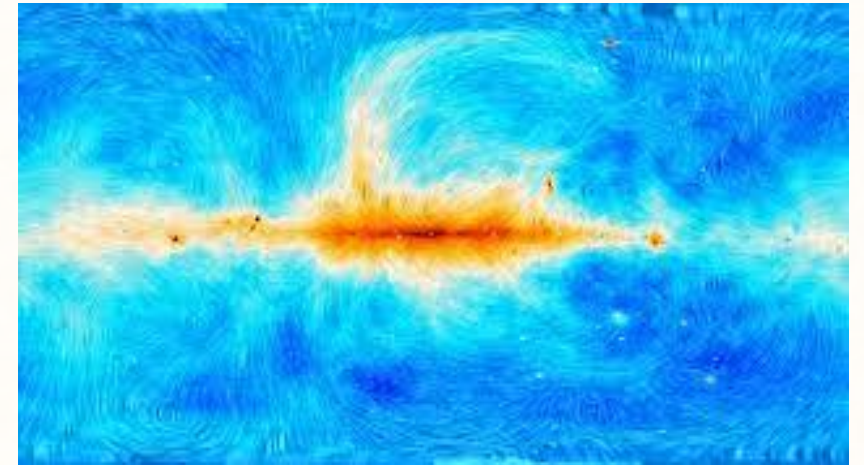
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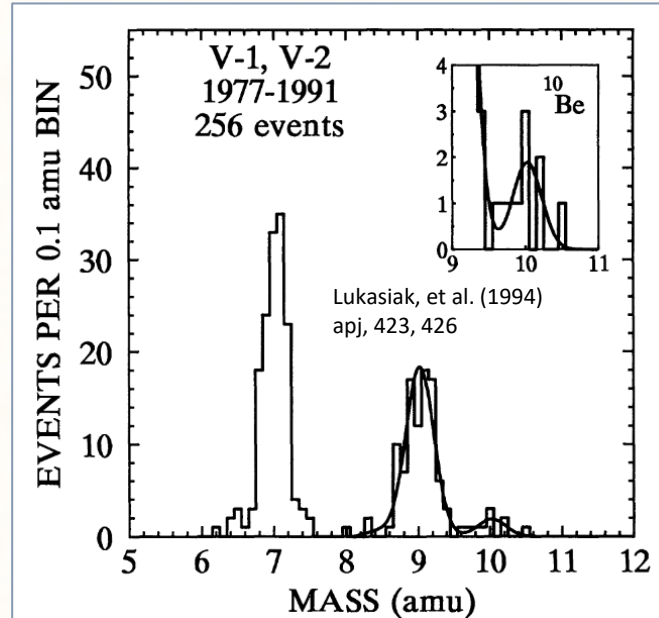
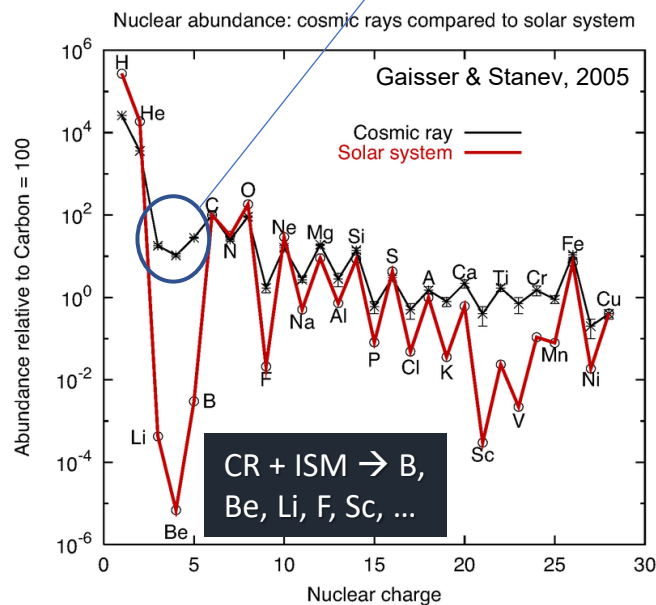


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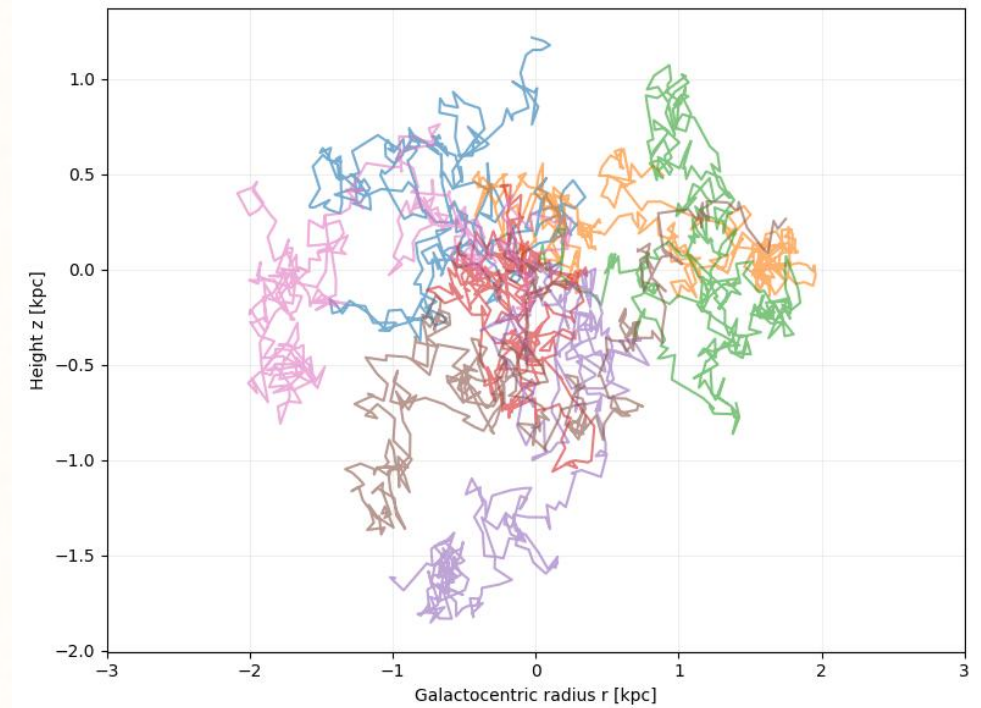
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Turbulence (interaction with eddies) gets CRs confined within the Galaxy



Resonant interactions with plasma waves

The magnetic Lorentz force is

$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}$$

If $\mathbf{v} \perp \mathbf{B}$, its magnitude is

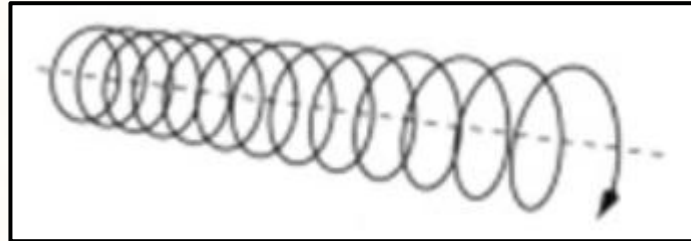
$$F = qvB$$

For circular motion the required centripetal force is

$$F = \frac{\gamma m v^2}{r_L}$$

$$F = \frac{pv}{r_L}$$

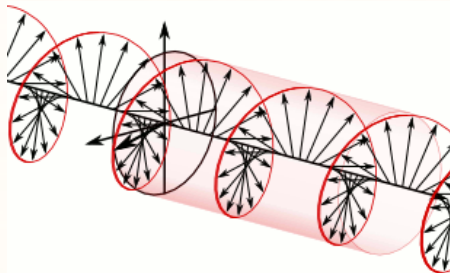
Gyromotion of a cosmic ray



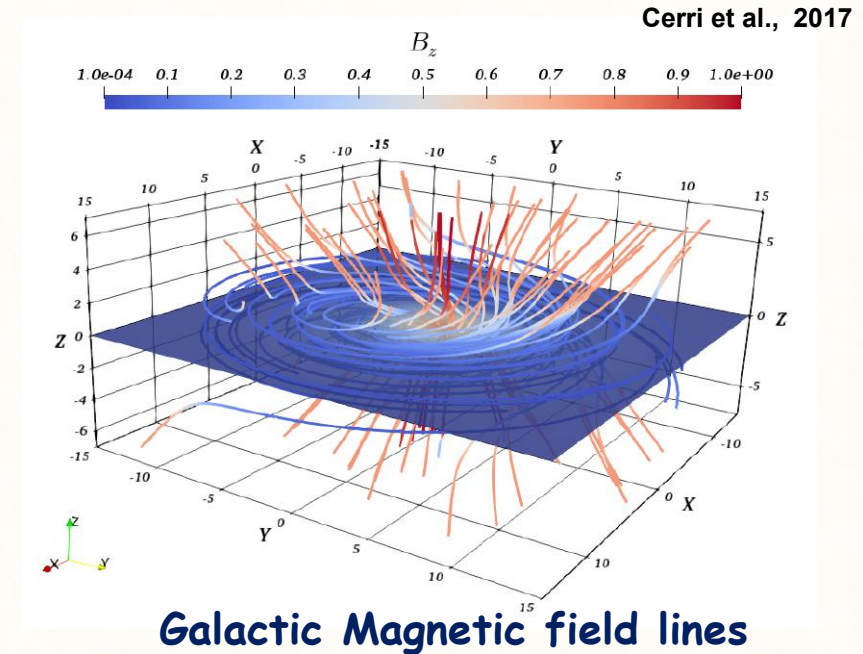
Larmor radius

$$r_L = \frac{E}{ZeBc}$$

Plasma instabilities (waves) also move along magnetic field lines -- and they **are circularly polarized**!



Resonant scattering produces that only particles with certain energy can interact with certain waves

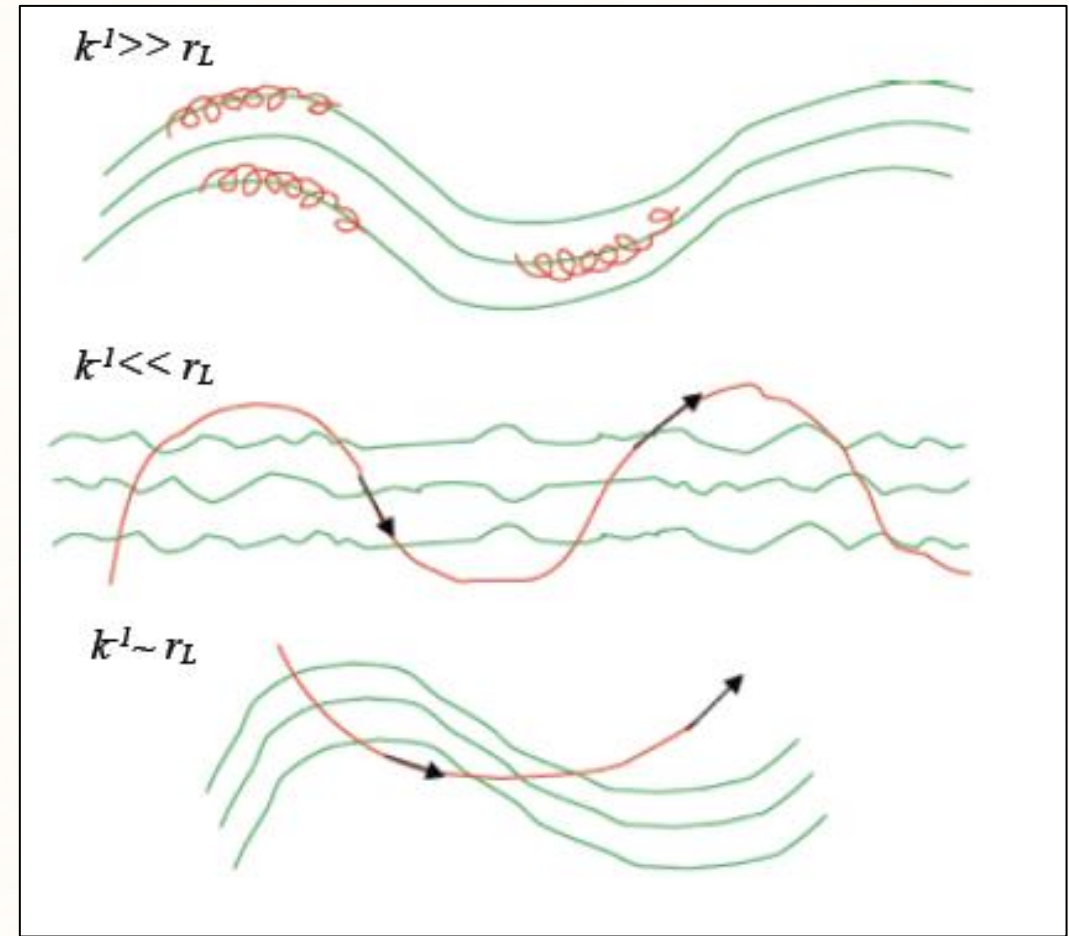
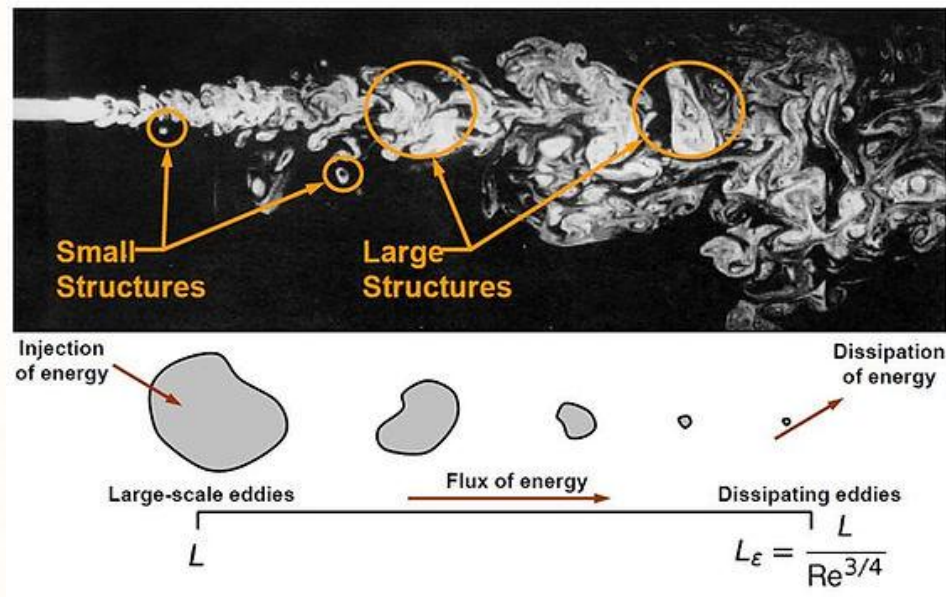


$$r_L \approx 1.08 \text{ kpc} \left(\frac{E}{1 \text{ PeV}} \right) \left(\frac{1}{Z} \right) \left(\frac{1 \mu\text{G}}{B} \right)$$

Resonant interactions with plasma waves

These interactions directly connect (or “couple”) cosmic rays with the interstellar medium and its turbulence!

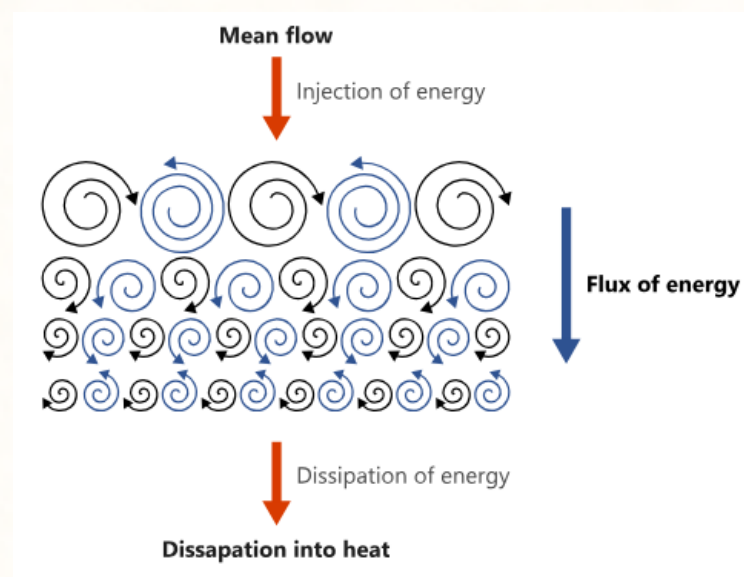
Therefore, the time of cosmic rays spend in the Galaxy, and the amount of secondary cosmic rays is directly related to the plasma conditions



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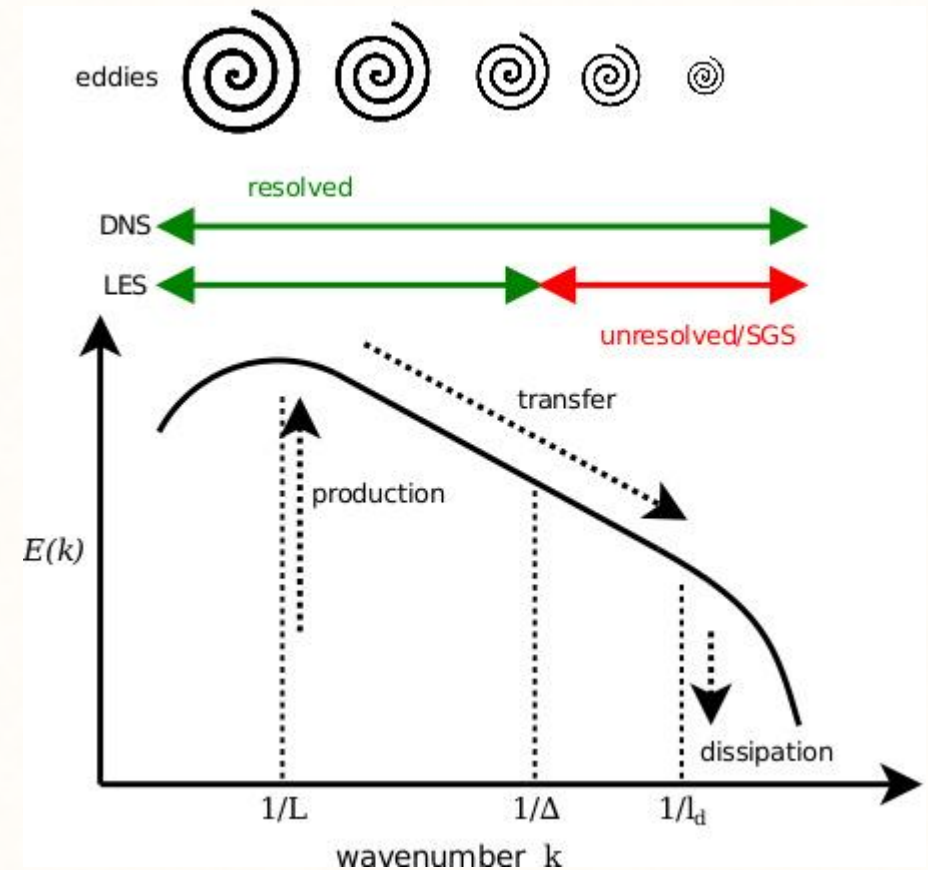
Therefore, the time of cosmic rays spend in the Galaxy, and the amount of secondary cosmic rays is directly related to the plasma conditions



This provides a **power-law relation for the interaction time**, proportional to the time that CRs stay in the Galaxy!

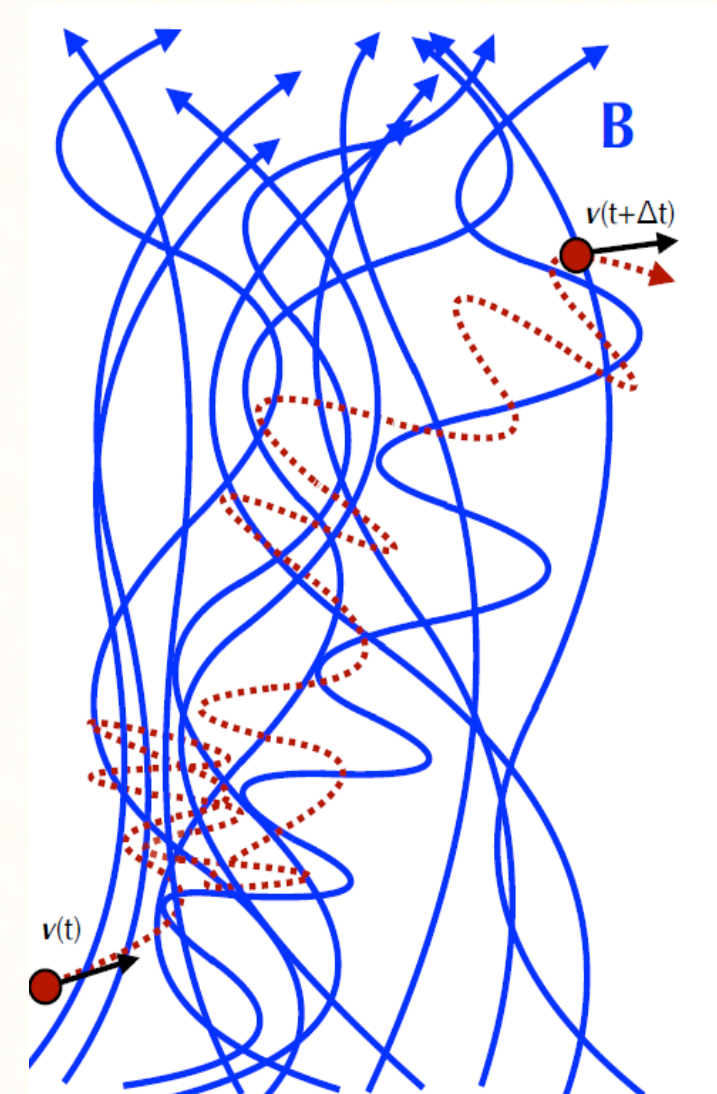
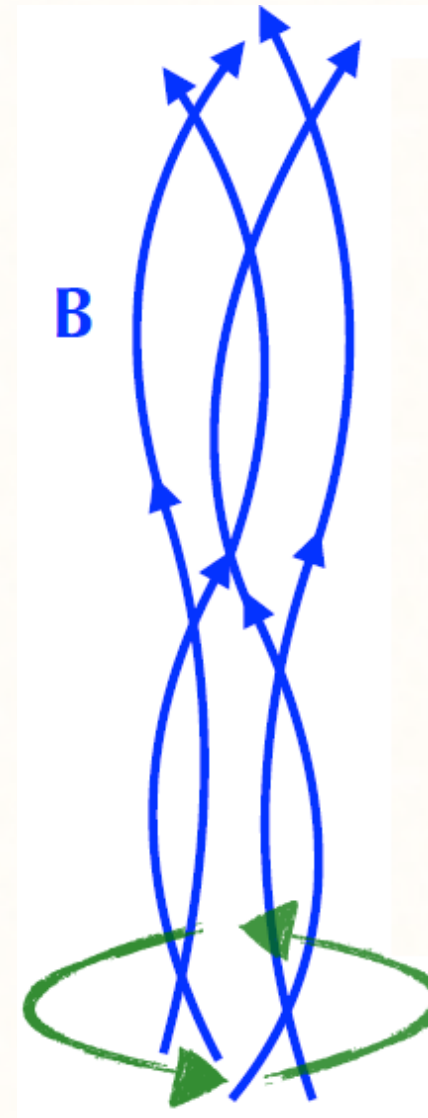
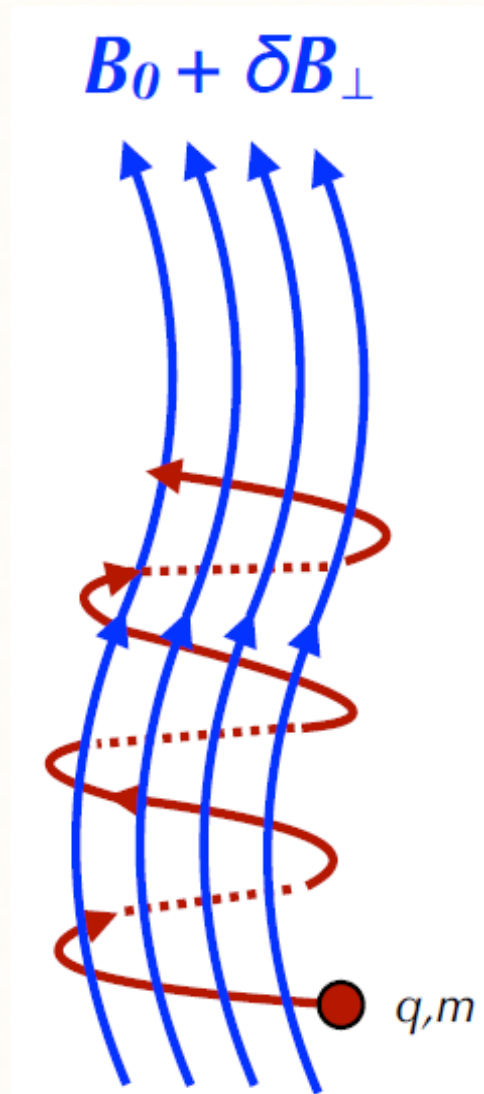
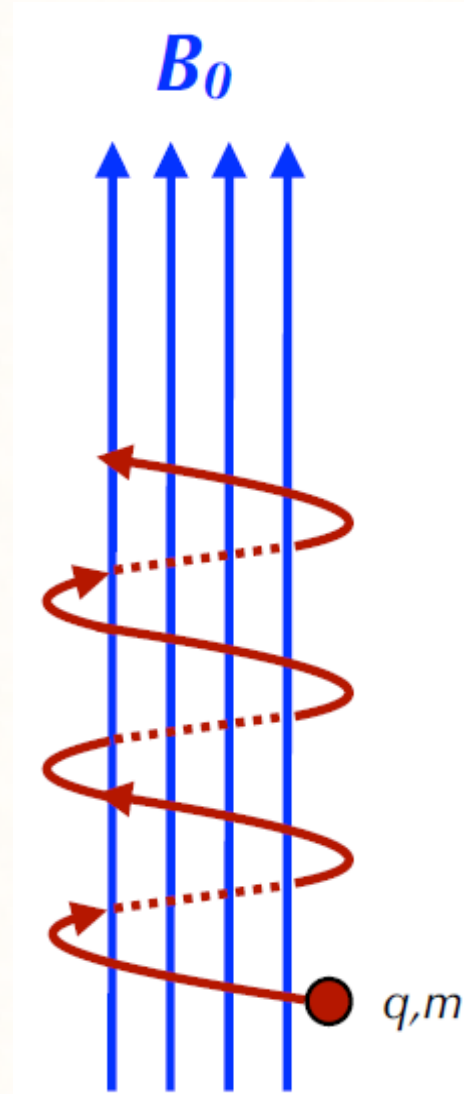
$$\tau_{\text{scat}} \propto r_L^\delta \propto \left(\frac{E}{ZB} \right)^\delta$$

The key point: The size of these plasma instabilities (eddies) are distributed as a power-law



CR interaction with the ISM

**Random plasma waves
isotropize the CR directions**



Resonant interactions lead to power-laws

CR observed flux: $J \propto E^{-\gamma}$; $\gamma \sim 2.7$

Injection from SNRs: $Q \propto E^{-\alpha}$; $\alpha \sim 2-2.3$

$$\cancel{\frac{dN_i}{dt}} = Q_i(E) - \frac{N_i}{\tau^{esc}} \quad (\text{Leaky box approx.})$$

Steady solution $N_i = Q_i \tau^{esc}$ $\longrightarrow$ Confinement time must be: $\tau^{esc} \propto E^{-\delta}$; $\delta \sim 0.5$

Solving the equation $\frac{d\vec{p}}{dt} = q \frac{\vec{v} \times (\vec{B} + \delta \vec{B})}{c}$ leads to spectra $\propto (\delta B/B)^2 \propto k^{-\lambda}$

The power spectrum of MHD turbulence follow a power law in wavenumber (k) and is able to explain the propagation of CRs as a diffusive transport in the Galaxy with $0.3 < \delta < 0.6$

Confinement time → Diffusive transport

- **This random motion can be described as a diffusion motion!**

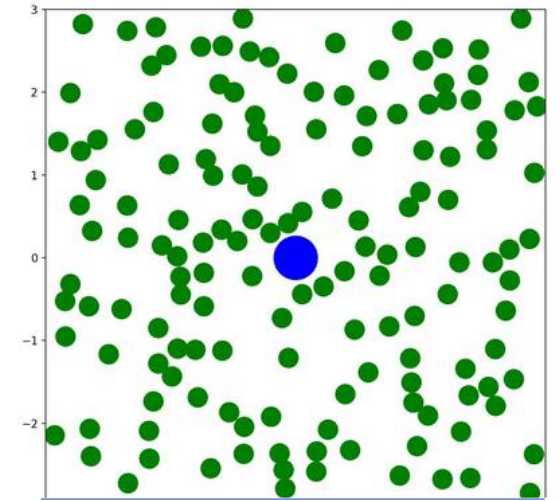
Here, the diffusion coefficient describes the effective interaction with plasma waves and quantify how quickly these particles move from their origin (i.e. their sources of injection)

- In this way, cosmic rays propagate through the Galaxy by **scattering off magnetic turbulence**. These randomizes their trajectories and make them cross the Galactic Plane many times.
- The diffusion coefficient is inversely proportional to the diffusion time (or directly proportional to the scattering time)

$$\tau \sim \frac{L^2}{D} \propto r_L^{-\delta}$$



$$D(E) \sim D_0 \left(\frac{E}{E_0} \right)^\delta$$



Units of the diffusion coefficient:

$$cm^2/s$$

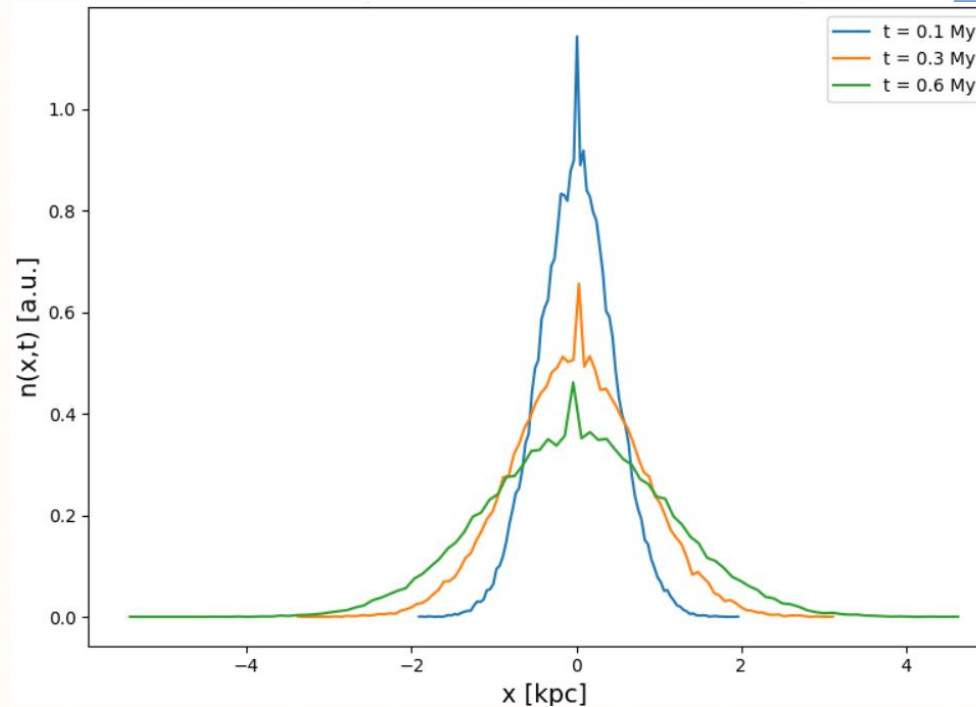
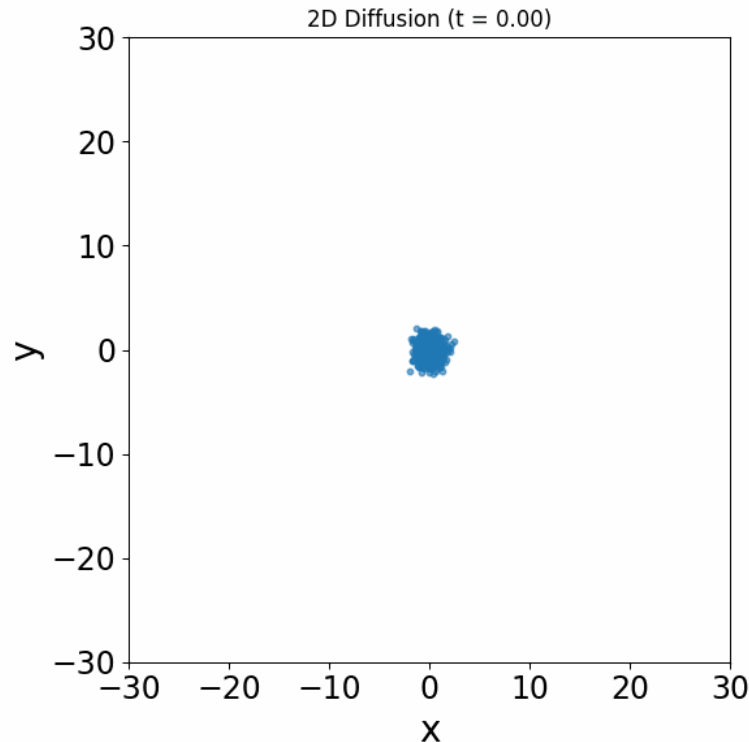
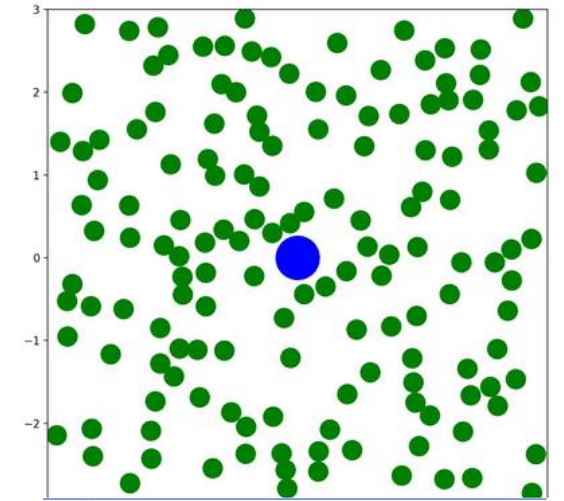
Typical values for D_0 at 1 GeV:

$$10^{28} cm^2/s \sim 0.03 kpc^2/Myr$$

Confinement time → Diffusive transport

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Here, the diffusion coefficient describes the effective interaction with plasma waves and quantify how quickly these particles move from their origin (i.e. their sources of injection)



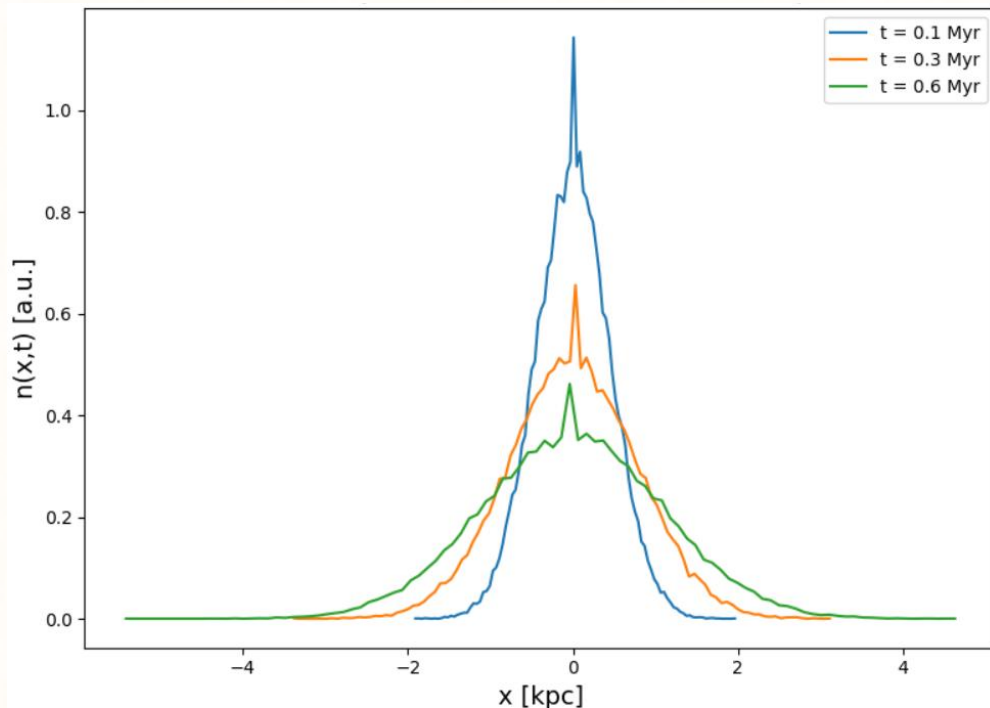
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Units of the D: cm^2/s

Confinement time → Diffusive transport

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$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$

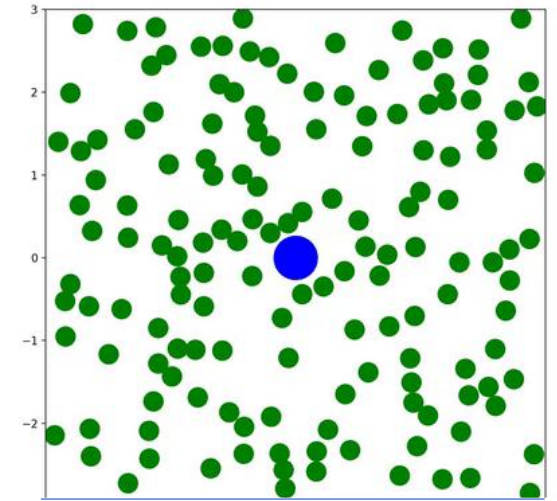
$$n(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

Mean

$$\langle x \rangle = 0$$

Variance

$$\langle x^2 \rangle = 2Dt$$



$$D(E) \sim D_0 \left(\frac{E}{E_0} \right)^\delta$$

Units of the D: cm^2/s

Confinement time → Diffusive transport

- **This random motion can be described as a diffusion motion!**

Here, the diffusion coefficient describes the effective interaction with plasma waves and quantify how quickly these particles move from their origin (i.e. their sources of injection)

- The diffusion equation that we should solve becomes very complicated when including other relevant effects. The minimal extension of the diffusion equation for cosmic rays will include a **source term**

- The source term must encode the power-law particle injection and the spatial distribution that supernova remnants follow:

$$Q(x, t, E) = S(E) \cdot L(t) \cdot R(x)$$

- As we'll see this will get very complicated, and no analytical solution can be generally obtained. Therefore, **we must rely on numerical solutions**

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2} + Q(x, t)$$

$$S(E) = S_0 \left(\frac{E}{E_0} \right)^{-\gamma}$$

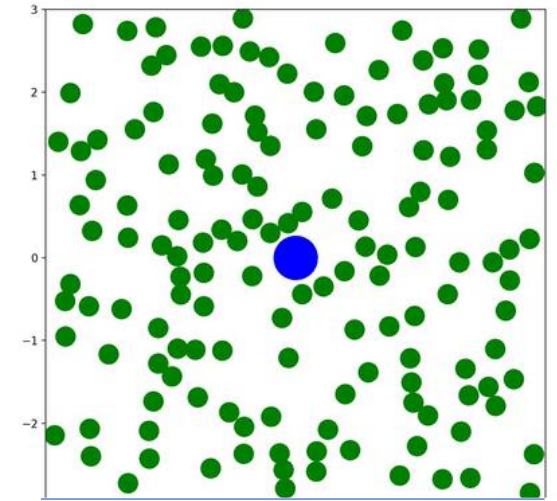
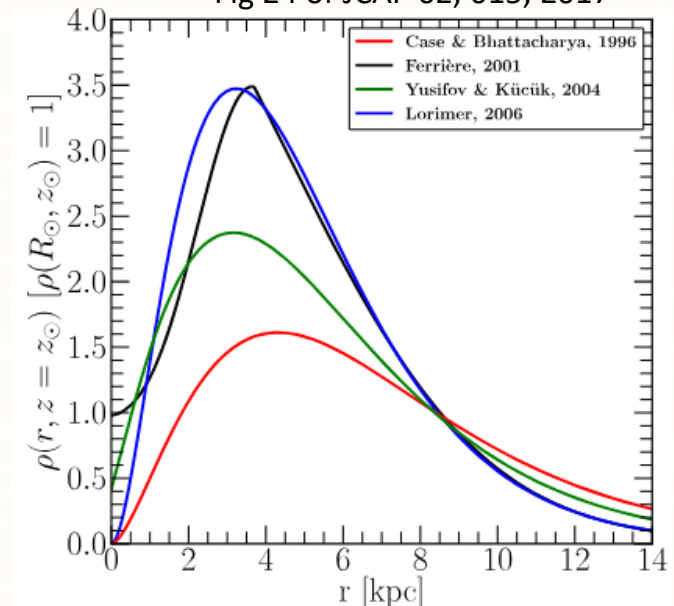


Fig 24 of JCAP 02, 015, 2017



Numerical solution to the diffusion equation

Discretization

Divide time in time steps

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$

Divide space as a grid of points.

$$\frac{\partial n}{\partial t} \approx \frac{n_i^{t+\Delta t} - n_i^t}{\Delta t}$$

$$\frac{\partial^2 n}{\partial x^2} \approx \frac{n_{i+1}^t - 2n_i^t + n_{i-1}^t}{\Delta x^2}$$

Explicit Forward Euler

$$\Delta t \leq \frac{\Delta x^2}{2D}$$

$$n_i^{t+\Delta t} = n_i^t + D \frac{\Delta t}{\Delta x^2} (n_{i+1}^t - 2n_i^t + n_{i-1}^t)$$

$$n_i^{t+\Delta t} = n_i^t + D \frac{\Delta t}{\Delta x^2} (n_{i+1}^t - 2n_i^t + n_{i-1}^t) + \Delta t Q_i$$

Numerical solution to the diffusion equation

Crank-Nicolson

Averages when doing spatial derivatives

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$

$$\frac{n_i^{n+1} - n_i^n}{\Delta t} = \frac{D}{2} \left[\frac{n_{i+1}^n - 2n_i^n + n_{i-1}^n}{\Delta x^2} + \frac{n_{i+1}^{n+1} - 2n_i^{n+1} + n_{i-1}^{n+1}}{\Delta x^2} \right]$$

$$-\frac{r}{2}n_{i-1}^{n+1} + (1+r)n_i^{n+1} - \frac{r}{2}n_{i+1}^{n+1} = \frac{r}{2}n_{i-1}^n + (1-r)n_i^n + \frac{r}{2}n_{i+1}^n$$

This method is “unconditionally stable”.

The best way to solve it is setting up a tridiagonal matrix and finding its inverse

$$A \mathbf{n}^{n+1} = \mathbf{b}, \quad A = \begin{bmatrix} 1+r & -r/2 & 0 & \cdots & 0 \\ -r/2 & 1+r & -r/2 & \cdots & 0 \\ 0 & -r/2 & 1+r & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & -r/2 \\ 0 & 0 & 0 & -r/2 & 1+r \end{bmatrix}$$

TO SUMMARIZE...

Introduction to Galactic Cosmic Rays

- Cosmic rays are charged particles from the outer space that are observed over a wide range of energies, and composed primarily of protons, with smaller contributions of heavier nuclei and electrons
- Most of these cosmic rays have a Galactic origin. In particular, they are believed to be ISM particles that are accelerated in shocks of astrophysical sources like supernova remnants
- Cosmic rays propagate through the Galaxy following a diffusive motion driven by their scattering with fluctuations of the Galactic magnetic field (plasma turbulence)
- Given the complexity of the processes that cosmic rays undergo in the Galaxy, numerical approaches are necessary to model their spectra and distribution